

The Method 1.6 — Spectra Compendium

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THE METHOD 1.6 — SPECTRA COMPENDIUM

The index of Rydberg channels, and a value for every cell of it.

This compendium supplies the book with coordinates. Its subject is Λ spectra; its output is a defect for every channel of every element, graded by how the value was got. *The register carries the working record — the corrections, the reversals, and what remains open. This carries the answers.*

0 · THE COORDINATE SUPPLY

What a coordinate is

$(Z, \text{charge}, \ell, 2S+1) \rightarrow \delta$, with a grade, a source and a bound

Z the atomic number · **charge** the ionisation stage, 1 for the neutral · ℓ the Rydberg orbital · **2S+1** the multiplicity, which Hund's rule on the core derives from the ground state and does not require measuring.

δ is the quantum defect: $E = -Z_c^2 R / (n - \delta)^2$, so a channel's whole Rydberg series follows from one number and the core charge.

DECLARATION — the stored δ is the MEDIAN over a cell's members (T2, closed). This was measured, not chosen: of 433 series whose member count reproduces exactly from raw, the stored value equals the median in 430 (99.3%); the 18 mean-agreements are series where median and mean coincide and distinguish nothing (R 1673, 1675, enlarged sample R 1679). The mean and asymptotic perspectives are carried alongside per cell, as perspectives of one definition (T6, M's ruling, session 1.7.3).

The grades

grade	rows	what it means
exact	929	forced by symmetry — $\delta = 0$ when one electron faces a bare nucleus
measured	358	fitted to captured levels against an ionisation limit
computed	103,545	supplied by the channel equation, within the domain stated below

Nothing is ungraded. A value with no provenance is not a coordinate.

The witness

*Register 1578. This table used to grade cells VERIFIED, POSSIBLE and IMPROBABLE. **Impossible is a judgement about the future, not a fact about the record** — and register 1287 had already established the rule it breaks: an index whose coordinates mix the OBJECT with the OBSERVER cannot close. What replaces it is a fact and a named obstacle.*

	cells	
the index admits	104,832	every channel of every

	cells	
witnessed	358	element reality has been consulted for this cell
unwitnessed	104,474	it has not — see the bound

0.341% of the index has been checked against reality. Every other cell is theory, and the compendium should be read that way.

The bounds, witnessed and unwitnessed

Not probabilities. Named obstacles, each of which may or may not be lifted.

cells	the obstacle
28,526	series unresolved above ng in any published analysis
24,312	no long-lived isotope
17,626	not keyable: no single 2S+1 (hole+electron or multi-valence)
11,605	open-shell core, 16 parents
9,756	open-shell core, 3 parents
5,280	open-shell core, 119 parents
2,691	no analysis located at this charge (NOT a bound on existence)
1,760	no nuclide synthesised; theoretically admitted — Janet left-step, 8s(2) closes the eighth period at element 120
1,744	no nuclide synthesised; theoretically admitted — Janet left-step, 8s(1) opens element 119
929	derived by symmetry; no measurement required
240	none — separable series, simply not yet measured
6	limit 41449.451; one electron outside a closed shell
5	no primordial isotope (R 1587)
5	limit 31406.4677325; one electron outside a closed shell
3	limit 66928.04; one electron outside a closed shell
3	limit 49266.66; one electron outside a closed shell
2	limit 48278.48; one electron outside a closed shell

cells	the obstacle
2	limit 48387.634; one electron outside a closed shell
2	limit 32848.872; one electron outside a closed shell
1	limit 46670.107; one electron outside a closed shell
1	limit 43762.6; one electron outside a closed shell

The charge bound is the one to distrust. It reads *no analysis located*, not *no analysis exists* — and register 1577 found twelve MEASURED cells above it in this index's own file (Ti XI, Fe XV, Fe XVI), with Theodosiou 1986 covering all positive ions of all atoms to $Z = 37$.

Accuracy figures in this compendium are quoted against the WITNESSED-plus-possible set, not against the whole index.

The bound, which needs no measurement at all

$$B = \min(p, n_o - \ell - 1)$$

with **p** the core's orbital count at that ℓ and **n_o** the first Pauli-allowed principal number, both read from the ground-state configuration. **floor(6) ≤ B holds on every measured channel without exception**, and B travels with every row of the table — so the book can bound a defect without consulting one.

The filler, and its domain

Where no capture reaches, the channel equation supplies the value:

$$\begin{aligned} p > 0 : \delta &= 0.3772 \cdot p^e(N_e) \cdot N_e^k \cdot \ln(c+1)/c \\ p = 0 : \delta &= 0.5415 \cdot C(Z) \cdot (N_e-1)/N_e \cdot N_e^k \cdot \ln(c+1)/c \end{aligned}$$

$$e(N_e) = 0.8297 - 0.0900 \cdot \ln N_e \quad k = 0.4942$$

$$N_e = Z - \text{charge} + 1$$

$$C(Z) = \text{the collapse coordinate across the Janet block boundary}$$

Domain: $Z \leq 92$, charge ≤ 10 , $\ell \leq 4$, a single-parent core. Fitted there on 277 channels at rms 0.1329 and R^2 0.9747, anchored at $Z = 2$ and $Z = 90$. *Every value outside the domain is unwitnessed, and the bound column names why.*

The table

The index is a rectangle, and the file says so: 13,104 keys of (Z, charge, 2S+1), each carried across all eight values of ℓ , 104,832 cells in all. Nothing is missing from it and nothing is chosen — what varies from row to row below is not how many cells exist but how they were got.

ℓ	cells	exact	measured	computed	δ , low to high
0 s	13,104	115	92	12,897	0.0000 ...

ℓ	cells	exact	measured	computed	δ , low to high
					6.0207
1 p	13,104	115	81	12,908	−0.0144 ... 5.6604
2 d	13,104	115	89	12,900	−0.1152 ... 4.8123
3 f	13,104	115	59	12,930	−0.2367 ... 3.6462
4 g	13,104	115	36	12,953	−0.0401 ... 0.0381
5 h	13,104	118	1	12,985	0.0000 ... 0.0001
6 i	13,104	118	0	12,986	0.0000 ... 0.0000
7 k	13,104	118	0	12,986	0.0000 ... 0.0000
	104,832	929	358	103,545	

The measured cells and the witnessed cells are the same 358 — not two counts that agree, the same set, tested cell by cell. A cell is measured exactly when reality was consulted for it, so the grade column and the witness column answer one question twice.

Two readings a reader should take from the shape. The first: the observation runs out at $\ell = 5$. One cell above g has ever been measured, and none above h ; everything the compendium says about high- ℓ channels is the channel equation speaking, and δ narrows as ℓ rises — the s channels span six units, the g channels a twelfth of one — which is the core's reach falling away, and it is why the Pauli bound of δ The bound, which needs no measurement at all costs nothing to state at high ℓ and everything at low.

Every figure in this table is read from the coordinate file set out in § The file, and each is checked against that file.

The file

COORDINATES-2.13 (CSV, 104,832 rows, one per admitted cell) is the coordinate index itself, and is delivered with this compendium as its data companion. Everything in Parts 0–II is read from it at build; the file is the object, the pages are the reading. A cell is cited from any volume as **COORD(Z, charge, ℓ , 2S+1)** — the four coordinates are the key and are unique in the file.

column	alphabet	meaning
Z	1 ... 120	the element; 119 and 120 admitted on Janet's authority, register 1578's bound column saying so
charge	1 ... Z	the ion; charge $\leq Z$ is the triangle, 7,260 (Z, charge)

column	alphabet	meaning
		pairs
l	0 ... 7	the running electron's ℓ
mult	1 ... 9	2S+1; the set allowed at a pair is a function of the electron count Z – charge alone, holding on all 7,260 pairs
delta	–0.2367 ... 6.0207	the quantum defect held for the channel
grade	exact · measured · computed	929 · 358 · 103,545 — one-electron cells are exact by symmetry
source	six provenances, 32 strings	the rule that computed the value, or the compilation and species it was taken from; the strings run to thirty-two because each NIST retrieval carries its own session
B	0.0000 ... 7.0000	the Pauli bound on δ , read from the ground state; nonzero on 36,917 cells, and integer-valued on all but twenty-five of them
witness	witnessed · unwitnessed	358 · 104,474 — reality consulted, or not
bound	22 distinct	the named obstacle — on every unwitnessed cell and on 25 witnessed ones (§ <i>The bounds, witnessed and unwitnessed</i>)

The six provenances, and why the column shows thirty-two strings. *Every cell says where its value came from, and the strings fall into six kinds: the channel equation, 103,545 cells · one electron, $\delta = 0$ by symmetry, 929 · captured levels, 325 · NIST ASD retrieved 2026-08-14, 25 · three species read from their own level files at register 1627, 3 · Theodosiou, Manson and Inokuti 1986, 5. They sum to 104,832. The thirty-two distinct strings are not thirty-two compilations: twenty-five of them are the same NIST retrieval, each carrying the session that fetched it and the series it drew, so the count of strings measures provenance detail and not variety of source. The compilations behind the measured channels are named in § B.1 Sources.*

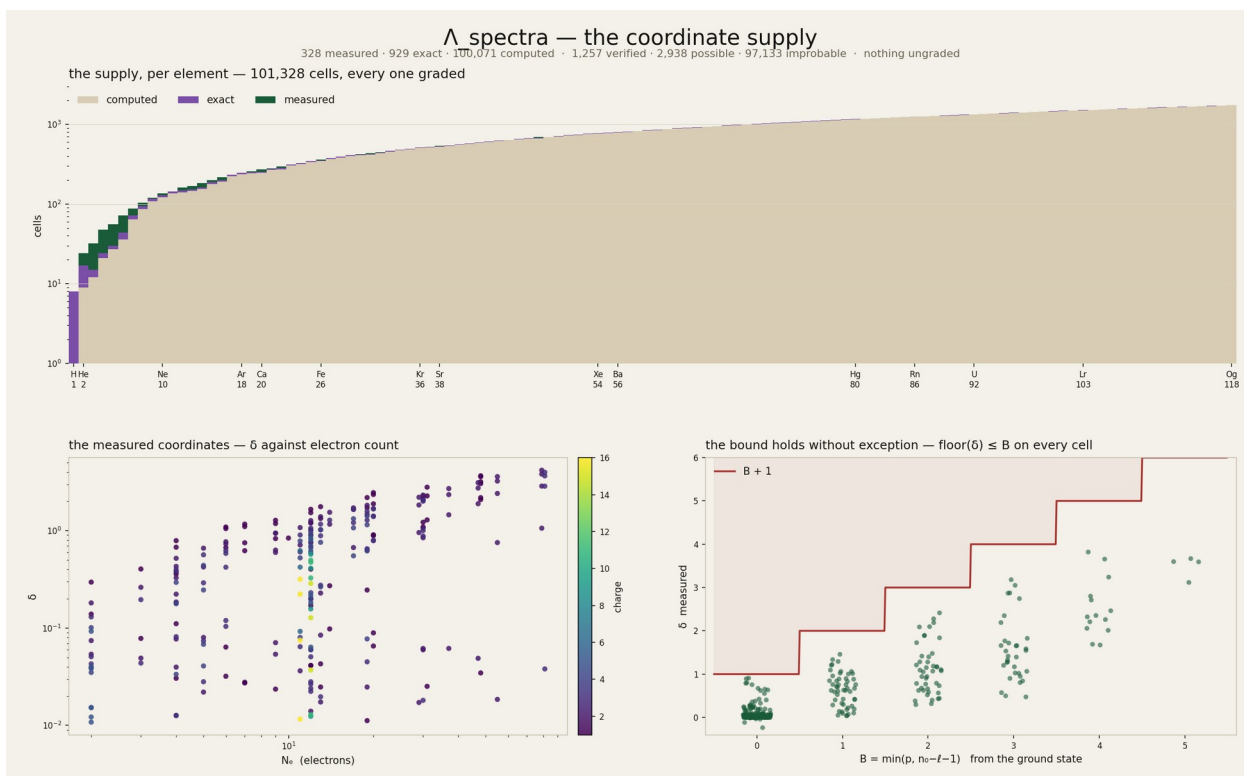
One row, as it stands in the file. *Sodium, singly charged nucleus, the s channel, doublet:*

Z=11 charge=1 l=0 mult=2 delta=1.34857 grade=measured
source=NIST ASD fetched 2026-08-14 (session 1.8 queue); NaI 2S n=3-20, 18 members; median defect

$B=0.02475$ witness=witnessed bound=limit 41449.451; one electron outside a closed shell

Read it as COORD(11, 1, 0, 2): the quantum defect of the sodium s series is 1.34857, measured rather than computed, taken as the median over eighteen members of the $n = 3-20$ series, and the Pauli bound the ground state imposes on it is 0.02475. A cell carries its value, how the value was got, who supplied it, and what stands in the way of getting a better one.

The seven columns Part 0 once printed — Z charge l mult delta grade source B status — were the file at register 630; status (improbable / possible) became witness and bound at register 1578, and Z was extended from 118 to 120 at the entry that admitted 3,504 cells on Janet's authority. Every count on these pages is read from the file; none is transcribed.



Above: the supply per element, on a log scale — the measured and exact cells against the computed. Below left: every measured coordinate, δ against electron count, coloured by charge. Below right: the Pauli bound, which holds on every measured cell and is read from the ground state alone.

0 · THE DERIVED SUPPLY — Λ_{chain} 's candidate spectrum

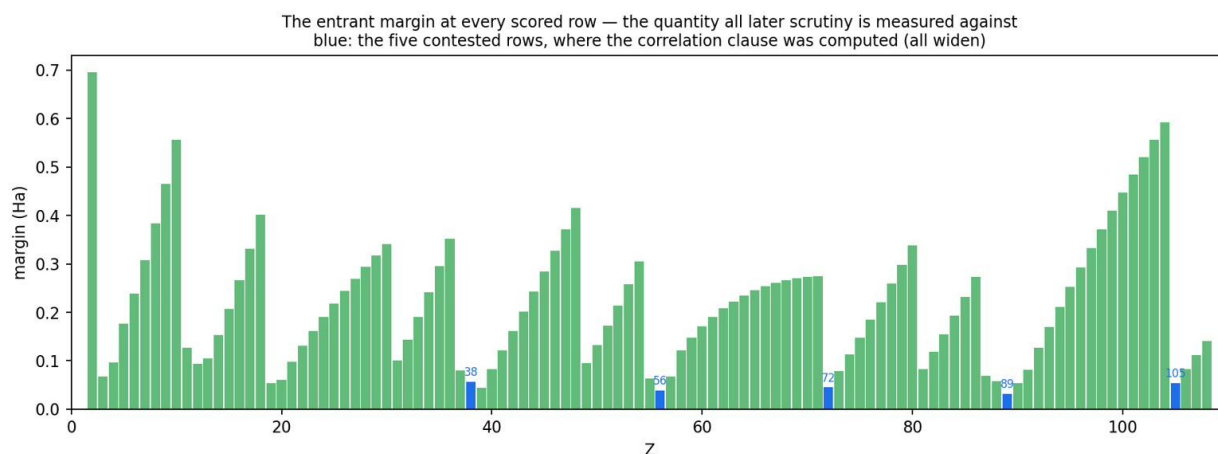
This compendium's founding supply runs observation-side: $(Z, \text{charge}, l, 2S+1) \rightarrow \delta$, graded by how the value was got. The Löwdin solution adds the equation-side supply:

at every element, a depth for every frontier channel, computed from the many-electron field with one constant and no measurement. The two supplies meet at the same atoms from opposite directions, and where one is silent the other speaks.

What a derived coordinate is

$(Z, n, \ell) \rightarrow D$, in hartree, with margin, provenance and score

Z the atomic number, neutral walk · **(n, ℓ)** any frontier channel offered at that step · **D** its converged one-channel depth in the frozen $V^{\{N-1\}}$ field · **margin** attached to the entrant row: the gap to the runner-up, the quantity every later scrutiny is measured against.



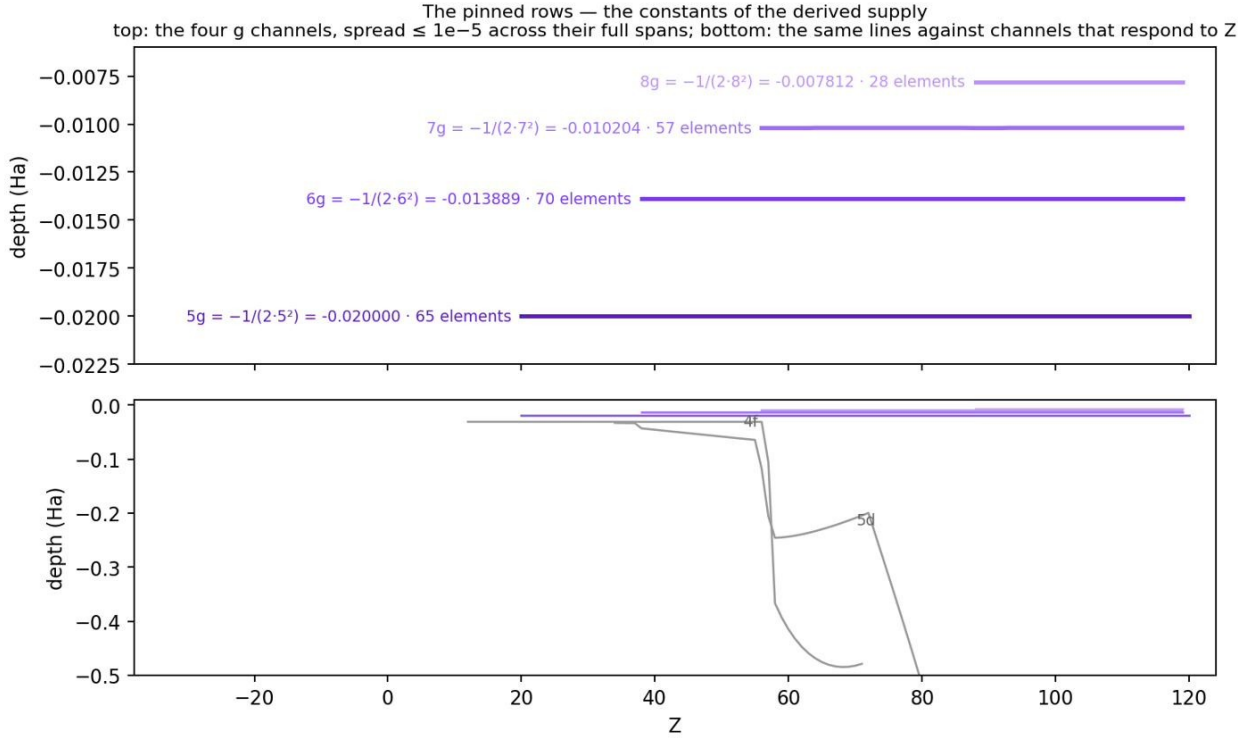
The entrant margin at every scored row. Blue: the five contested rows, where the correlation clause was computed and every competition widened.

The grades

grade	meaning	rows
SCORED	the entrant met the measured ground configuration	107 of 107, Z = 2-108
UNWITNESSED	no measurement exists to meet; published falsifiable	12, Z = 109-120
PINNED	the channel's depth is hydrogenic and Z-independent	every g row

UNWITNESSED carries this book's own sense: not undefected for want of trying — the rows await a measurement science has yet to provide (register 1712).

The pinned rows, stated as values



The pinned rows — the constants of the derived supply. Top: the four g channels, each at $-1/(2n^2)$ to storage precision, spread $\leq 10^{-5}$ across spans of 28 to 70 elements. Bottom: the same lines against the f and d channels that respond to Z.

5g: -0.020000 over 65 elements · 6g: -0.013889 over 70 · 7g: -0.010204 over 57 · 8g: -0.0078125 over 28 — each equal to $-1/(2n^2)$ to storage precision, spread $\leq 10^{-5}$ across the full span.

A channel whose value the nucleus cannot move needs no measurement and admits no fit. The compendium records these once, as constants of the walk, and the absence of a g series anywhere in the observation-side supply is thereby explained rather than merely noted (register 1704).

The unwitnessed twelve

Z	entrant	D_ent	margin
109-112	6d	$-0.326 \rightarrow -0.423$	$0.171 \rightarrow 0.264$
113-118	7p	$-0.160 \rightarrow -0.360$	$0.058 \rightarrow 0.210$
119, 120	8s	$-0.159, -0.189$	$0.097, 0.098$

Worst-case spin-orbit narrowing 0.083 Ha; every margin clears it. *When the spectra can be taken, these rows are scored like any others, and this table is where a miss would land.*

How the two supplies interlock

Where the observation supply goes silent, the derived supply names why. The f corridor of Chapter 34 has no test ($L = -\infty$ at the node floor); the derived supply puts the collapse condition at that exact address and decides La, Ac, Th there (register 1703). **Where the derived supply is unwitnessed, the observation supply states what witnessing would take.** And at the hundred-odd elements both supplies cover, they are independent routes to one order — *the strongest cross-check either one owns.*

I · THE INDEX

The four coordinates, and where each comes from

coordinate	alphabet	derived from
Z	1 ... 120	given; 119-120 admitted, unwitnessed
charge	1 ... Z	given
ℓ	0 ... 7	given
2S+1	Hund's rule on the core	the ground state

The multiplicity is not a free coordinate. Hund's first rule applied to the core's ground configuration gives the allowed values, exact on all 24 electron counts tested. Applying it as a constraint removes **247,248 cells** — **71%** of what the naive product would admit.

Which cells exist

A cell is admitted when $\text{charge} \leq Z$, at least one electron remains, the multiplicity is one Hund's rule allows, and an orbital of that ℓ is available. **Every cell the index admits is physically possible:** the forbidden ones were excluded at construction, so the survey holds none.

Which cells can be measured

restriction	cells
the index admits	104,832
$Z \leq 92$ — naturally occurring	61,152
and $\text{charge} \leq 10$	11,416
and $\ell \leq 4$ — where series resolve	4,395
and a single-parent core	1,755

The parent-term wall

A closed-shell core has one parent term and gives one Rydberg series per ℓ . An open-shell core gives one series per parent, all interleaved, converging on different limits.

Fe IV's $3d^4$ core carries sixteen LS terms. Its published levels show thirteen of them across **24 distinct (parent, ℓ , term) series, every one with a single member**. Fe IV has about a thousand analysed levels and no extractable defect: *the levels are identified and the series are not separable*.

This is why the compendium holds Ne-like and Na-like ions at charge 15 and 16 and no open-shell ion above charge 6. An open-shell core does not produce the object a quantum-defect index holds.

The Janet collapse

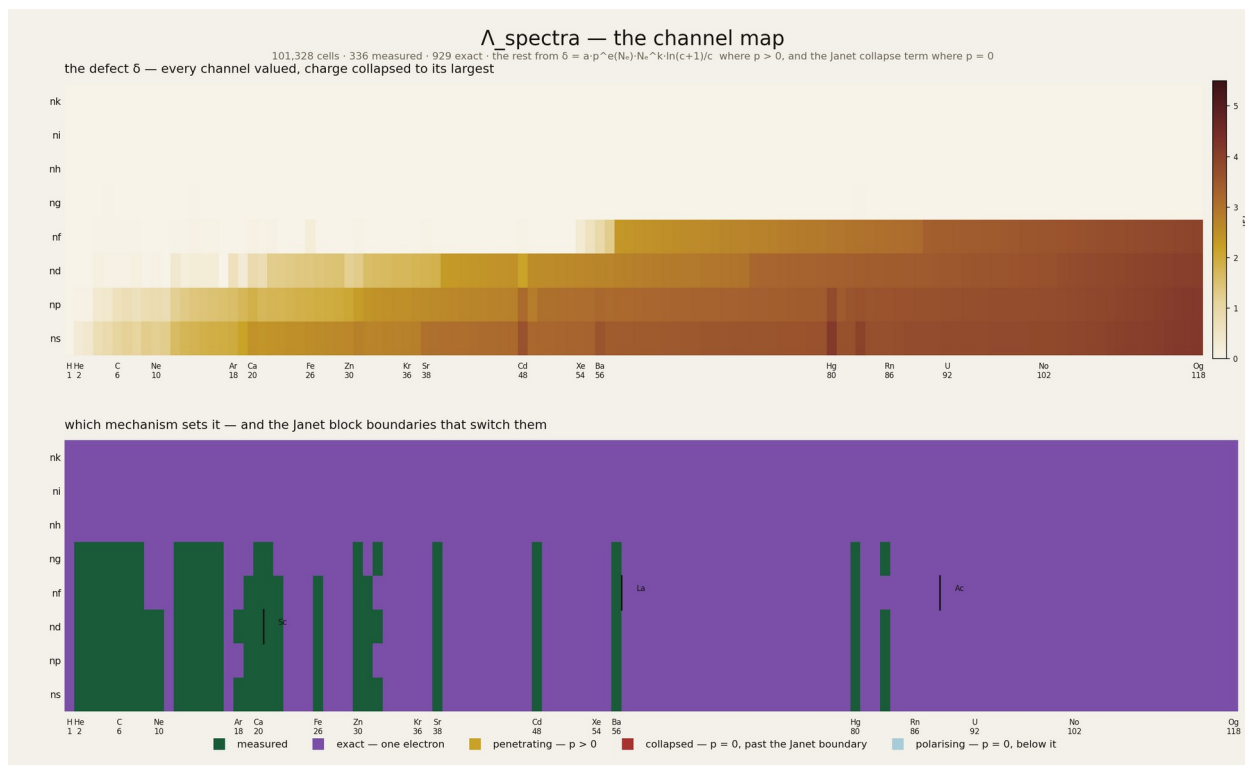
Two channels can both have $p = 0$ — no core orbitals of their own ℓ — and differ by a factor of ten. Ti IV's nd is **0.6202**; Sr II's nf is **0.0618**.

What separates them is **orbital collapse**, and its threshold is a Janet block boundary exactly:

orbital	$n+\ell$	block opens at	element	collapse threshold
3d	5	Z = 21	Sc	21
4f	7	Z = 57	La	57
5f	8	Z = 89	Ac	89

Across 116 cells with $p = 0$ at $\ell = 2$ or 3: **collapsed median 0.637, uncollapsed 0.036**, U-test $p = 9.8 \times 10^{-4}$. The transition is rapid rather than sharp — the largest defects are the atoms *approaching* a boundary: Ca I nd = 0.908 at $Z = 20$ against a threshold of 21, and Ba II nf = 0.756 at 56 against 57.

The collapse coordinate is read from the periodic table, not fitted, and it is why a defect at $\ell = 2$ can be large in one element and negligible in its neighbour.



Every cell valued. The upper panel is δ itself, charge collapsed to its largest; the lower is which mechanism sets it, with the three Janet boundaries marked.

II · THE CHANNELS

Every captured series, with its fit. These are the measured rows of the coordinate table.

Three marks state a limitation. An **asterisk** on the species marks a two-member channel: two points give a defect and one consistency check but cannot detect a perturbation. A **dagger** on the n-range marks an internal gap. And the **limit column** names how the ionisation limit was obtained — *published* where a source quotes it, *theoretical* where it is exact by construction, *constructed* where this work built it by summing two spectra. A *channel resting on a limit this work computed is not a measurement against an independent standard, and the row says so*. The **bracket column** reads m/k — of k cells with true measured neighbours on both sides, m pass the sealed test under ruling 26 (strict membership, quotation-floor guard, §22.5 admissibility; register 1763) — or no-triple where the row's members contain no three consecutive n, or untested where the run's data conditions are not met (the count paragraph below the table gives the split).

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Al I	3s ² nf ² F°	4-55	52	50	50/50	4.0-55.0	+0.0429	0.0552	1	48,278.480
Al I	3s ² nd ² D	3-34	32	30	30/30	2.6-33.7	+0.0248	0.6297	1	48,278.480
Al I	3s ² ns ² S	4-16	13	11	11/11	2.2-14.2	+1.7670	0.0537	1	48,278.480
Al I	3s ² np ² P°	3-7	5	3	3/3	1.5-5.7	+1.3365	0.2080	1	48,278.480
Al II	3sns ¹ S	4-16	13	11	11/11	2.8-14.8	+1.2028	0.0147	2	151,862.500
Al II	3sns ³ S	4-13	10	8	8/8	2.7-11.7	+1.2711	0.0474	2	151,862.500
Al II	3snd ¹ D	3-11	9	7	7/7	3.2-10.9	+0.0603	0.3897	2	151,862.500
Al II	3sng	5-13	9	7	7/7	5.0-13.0	+0.0212	0.0070	2	151,862.500

The Method 1.6

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Al II	3snd ³ D	3-10	8	6	6/6	2.8-9.8	+0.2024	0.0096	2	151,862.500
Al II	3snf ³ F°	4-11	8	6	6/6	3.9-11.0	+0.0140	0.4088	2	151,862.500
Al II	3snp ³ P°	4-10	7	5	5/5	3.1-9.1	+0.8982	0.0571	2	151,862.500
Al III	nd 2D J=5/2	3-8	6	4	4/4	2.9-7.9	+0.0644	0.0071	3	229,445.710
Al III	nf 2F* J=5/2	4-9	6	4	4/4	4.0-9.0	+0.0044	0.0005	3	229,445.710
Al III	ng 2G J=7/2	5-9	5	3	2/3	5.0-9.0	+0.0010	0.0001	3	229,445.710
Al III	nh 2H* J=9/2	6-9	4	2	1/2	6.0-9.0	+0.0003	0.0000	3	229,445.710
Al III	np 2P* J=1/2	3-7	5	3	3/3	2.4-6.4	+0.6056	0.0128	3	229,445.710
Al III	ns 2S J=1/2	3-8	6	4	4/4	2.1-7.1	+0.9049	0.0098	3	229,445.710
Al IV *	nf 2[5/2] J=3	4-5	2	1	no-triple	4.0-5.0	+0.0349	0.0100	4	968,924.400
Al IV	ns 2[3/2]* J=2	3-5	3	1	0/1	2.2-4.2	+0.7638	0.0042	4	968,924.400
Ar I	nd 2[7/2]* J=4	3-6	4	2	2/2	2.8-5.6	+0.3516	0.0715	1	127,198.600
Ar I	np 2[1/2] J=1	4-7	4	2	1/2	2.2-5.2	+1.7855	0.0204	1	127,198.600
Ar I	np 2[3/2] J=1	4-7	4	2	1/2	2.3-5.3	+1.7107	0.0123	1	127,198.600
Ar I	np 2[5/2] J=3	4-7	4	2	1/2	2.2-5.2	+1.7437	0.0119	1	127,198.600
Ar I	ns 2[3/2]* J=1	4-7	4	2	1/2	1.8-4.8	+2.1701	0.0138	1	127,198.600
Ar I	ns 2[3/2]* J=2	4-7	4	2	1/2	1.8-4.8	+2.1886	0.0132	1	127,198.600
Ar II	(³ P)ns ⁴ P J=5/2	4-6	3	1	1/1	2.2-4.3	+1.7463	0.0477	2	222,848.300
Ar II	(³ P)ns ⁴ P J=3/2	4-6	3	1	1/1	2.2-4.3	+1.7246	0.0663	2	222,848.300
Ar II	(³ P)ns ⁴ P J=1/2	4-6	3	1	1/1	2.2-4.4	+1.6821	0.1507	2	222,848.300
Ar II	(³ P)ns ² P J=3/2	4-6	3	1	1/1	2.3-4.4	+1.6711	0.1084	2	222,848.300
Ar II	(³ P)ns ² P J=1/2	4-6	3	1	1/1	2.3-4.4	+1.6355	0.1526	2	222,848.300
Ar II	(³ P)nd ⁴ D J=7/2	3-5	3	1	1/1	2.2-4.4	+0.6930	0.1695	2	222,848.300
Ar II	(³ P)nd ⁴ D J=5/2	3-5	3	1	1/1	2.2-4.4	+0.6880	0.1753	2	222,848.300
Ar II	(³ P)nd ⁴ D J=3/2	3-5	3	1	1/1	2.2-4.4	+0.6786	0.1919	2	222,848.300
Ar II	(³ P)nd ⁴ D J=1/2	3-5	3	1	1/1	2.2-4.4	+0.6669	0.2155	2	222,848.300
Ar II	(³ P)nd ⁴ F J=9/2	3-5	3	1	1/1	2.3-4.4	+0.6121	0.0885	2	222,848.300
Ar II	ng 2[4] J=9/2	6-8	3	1	untested	6.0-8.0	+0.0089	0.0036	2	222,848.300
Ar II	ng 2[6] J=11/2	6-8	3	1	untested	6.0-8.0	+0.0035	0.0003	2	222,848.300
Ar II	ng 2[6] J=13/2	6-8	3	1	untested	6.0-8.0	+0.0035	0.0002	2	222,848.300
Ar II *	ng 2[2] J=3/2	6-7	2	1	untested	6.0-7.0	-0.0006	0.0002	2	222,848.300
Ar II *	ng 2[2] J=5/2	6-7	2	1	untested	6.0-7.0	-0.0006	0.0002	2	222,848.300
Ar II *	ng 2[3] J=5/2	6-7	2	1	untested	6.0-7.0	+0.0063	0.0001	2	222,848.300
Ar II *	ng 2[3] J=7/2	6-7	2	1	untested	6.0-7.0	+0.0063	0.0001	2	222,848.300
Ar II *	ng 2[4] J=7/2	6-7	2	1	untested	6.0-7.0	+0.0114	0.0001	2	222,848.300
Ar II *	ng 2[5] J=11/2	6-7	2	1	untested	6.0-7.0	+0.0119	0.0002	2	222,848.300
Ar II *	ng 2[5] J=9/2	6-7	2	1	untested	6.0-7.0	+0.0119	0.0002	2	222,848.300
Ar II *	nh 2[3]* J=5/2	6-7	2	1	untested	6.0-7.0	-0.0017	0.0001	2	222,848.300
Ar II *	nh 2[3]* J=7/2	6-7	2	1	untested	6.0-7.0	-0.0017	0.0001	2	222,848.300
Ar II *	nh 2[4]* J=7/2	6-7	2	1	untested	6.0-7.0	+0.0021	0.0001	2	222,848.300
Ar II *	nh 2[4]* J=9/2	6-7	2	1	untested	6.0-7.0	+0.0021	0.0001	2	222,848.300
Ar II *	nh 2[5]* J=11/2	6-7	2	1	untested	6.0-7.0	+0.0047	0.0001	2	222,848.300
Ar II *	nh 2[5]* J=9/2	6-7	2	1	untested	6.0-7.0	+0.0047	0.0001	2	222,848.300
Ar II *	nh 2[6]* J=11/2	6-7	2	1	untested	6.0-7.0	+0.0045	0.0001	2	222,848.300
Ar II *	nh 2[6]* J=13/2	6-7	2	1	untested	6.0-7.0	+0.0045	0.0001	2	222,848.300
Ar II *	nh 2[7]* J=13/2	6-7	2	1	untested	6.0-7.0	-0.0000	0.0001	2	222,848.300
Ar II *	nh 2[7]* J=15/2	6-7	2	1	untested	6.0-7.0	-0.0000	0.0001	2	222,848.300
Ar II *	ni 2[4] J=7/2	7-8	2	1	untested	7.0-8.0	-0.0009	0.0005	2	222,848.300
Ar II *	ni 2[4] J=9/2	7-8	2	1	untested	7.0-8.0	-0.0009	0.0005	2	222,848.300
Ar II *	ni 2[5] J=11/2	7-8	2	1	untested	7.0-8.0	+0.0012	0.0003	2	222,848.300
Ar II *	ni 2[5] J=9/2	7-8	2	1	untested	7.0-8.0	+0.0012	0.0003	2	222,848.300
Ar II *	ni 2[6] J=11/2	7-8	2	1	untested	7.0-8.0	+0.0025	0.0002	2	222,848.300
Ar II *	ni 2[6] J=13/2	7-8	2	1	untested	7.0-8.0	+0.0025	0.0002	2	222,848.300
Ar II *	ni 2[7] J=13/2	7-8	2	1	untested	7.0-8.0	+0.0023	0.0002	2	222,848.300
Ar II *	ni 2[7] J=15/2	7-8	2	1	untested	7.0-8.0	+0.0023	0.0002	2	222,848.300
Ar II *	ni 2[8] J=15/2	7-8	2	1	untested	7.0-8.0	-0.0000	0.0006	2	222,848.300
Ar II *	ni 2[8] J=17/2	7-8	2	1	untested	7.0-8.0	-0.0000	0.0006	2	222,848.300
Ar II	3s2.3p4.(3P).nd 4D J=7/2	3-5	3	1	untested	2.2-4.4	+0.6930	0.0749	2	222,848.300
Ar II	3s2.3p4.(3P).ns 4P J=1/2	4-6	3	1	untested	2.2-4.4	+1.6821	0.0615	2	222,848.300
Ar II	3s2.3p4.(3P).ns 4P J=3/2	4-6	3	1	untested	2.2-4.3	+1.7246	0.0283	2	222,848.300
Ar II	3s2.3p4.(3P).ns 4P J=5/2	4-6	3	1	untested	2.2-4.3	+1.7463	0.0203	2	222,848.300
B II	nd 1D J=2	3-7	5	3	2/3	3.0-7.1	-0.0323	0.0111	2	202,691.900
B II	nd 3D J=1	3-6	4	2	2/2	2.9-5.9	+0.0772	0.0138	2	202,691.900
B II	nf 3F* J=2	4-7	4	2	2/2	4.0-6.9	+0.0397	0.0132	2	202,691.900
B II	ng 3G J=3	5-7	3	1	1/1	5.0-7.1	-0.0401	0.0192	2	202,691.900
B II	np 1P* J=1	3-5	3	1	1/1	2.7-4.6	+0.3284	0.0668	2	202,691.900
B II	np 3P* J=0	3-6	4	2	2/2	2.7-5.8	+0.2231	0.0345	2	202,691.900
B II	ns 1S J=0	3-6	4	2	1/2	2.6-5.6	+0.4303	0.0172	2	202,691.900
B II	ns 3S J=1	3-7	5	3	2/3	2.5-6.5	+0.5211	0.0148	2	202,691.900
B III	nd 2D J=3/2	3-11	9	7	7/7	3.0-11.0	+0.0023	0.0002	3	305,930.800
B III	nf 2F* J=5/2	4-11	8	6	5/6	4.0-11.0	+0.0003	0.0000	3	305,930.800
B III	ng 2G J=7/2	5-10	6	4	3/4	5.0-10.0	+0.0001	0.0000	3	305,930.800
B III	nh 2H* J=9/2	6-10	5	3	3/3	6.0-10.0	+0.0000	0.0000	3	305,930.800
B III	np 2P* J=1/2	2-9	8	6	3/6	2.0-9.0	+0.0437	0.0007	3	305,930.800
B III	ns 2S J=1/2	2-9	8	6	6/6	1.8-8.8	+0.1959	0.0030	3	305,930.800
B IV	1s.nd 1D J=2	3-10	8	6	2/6	3.0-10.0	+0.0007	0.0002	4	2,091,995.451
B IV	1s.nd 3D J=1	3-10	8	6	3/6	3.0-10.0	+0.0021	0.0003	4	2,091,995.451
B IV	1s.nf 1F* J=3	4-9	6	4	3/4	4.0-9.0	+0.0004	0.0005	4	2,091,995.451
B IV	1s.ng 1G J=4	5-9	5	3	0/3	5.0-9.0	+0.0002	0.0004	4	2,091,995.451
B IV	1s.nh 1H* J=5	6-9	4	2	1/2	6.0-9.0	-0.0002	0.0002	4	2,091,995.451
B IV *	1s.ni 1I J=6	7-8	2	1	no-triple	7.0-8.0	-0.0003	0.0003	4	2,091,995.451
B IV	1s.np 1P* J=1	3-10	8	6	3/6	3.0-10.0	-0.0096	0.0003	4	2,091,995.451
B IV	1s.np 3P* J=0	3-10	8	6	6/6	3.0-10.0	+0.0350	0.0001	4	2,091,995.451
B IV	1s.ns 1S J=0	3-10	8	6	4/6	3.0-10.0	+0.0392	0.0004	4	2,091,995.451
B IV	1s.ns 3S J=1	3-10	8	6	6/6	2.9-9.9	+0.1011	0.0009	4	2,091,995.451
B V	nd 2D J=3/2	3-10	8	6	6/6	3.0-10.0	+0.0005	0.0002	5	2,744,111.380
B V	nf 2F* J=5/2	4-10	7	5	5/5	4.0-10.0	+0.0004	0.0002	5	2,744,111.380
B V	ng 2G J=7/2	5-9	5	3	3/3	5.0-9.0	+0.0003	0.0002	5	2,744,111.380
B V	nh 2H* J=9/2	6-9	4	2	2/2	6.0-9.0	+0.0003	0.0001	5	2,744,111.380
B V	ni 2I J=11/2	7-9	3	1	1/1	7.0-9.0	+0.0004	0.0001	5	2,744,111.380
B V *	nk 2K* J=13/2	8-9	2	1	no-triple	8.0-9.0	+0.0004	0.0001	5	2,744,111.380

The Method 1.6

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
B V	np 2P* J=1/2	3-10	8	6	6/6	3.0-10.0	+0.0008	0.0002	5	2,744,111.380
B V	ns 2S J=1/2	3-10	8	6	6/6	3.0-10.0	+0.0008	0.0002	5	2,744,111.380
Ba II	nd 2D J=3/2	5-25†	14	12	10/10	2.4-22.6	+2.4147	0.0516	2	80,687.900
Ba II	nf 2F* J=5/2	4-12	9	7	7/7	3.7-11.1	+0.7559	0.1645	2	80,687.900
Ba II	ng 2G J=7/2	5-10	6	4	4/4	5.0-10.0	+0.0185	0.0022	2	80,687.900
Ba II	np 2P* J=1/2	6-12	7	5	5/5	2.7-8.8	+3.2408	0.0287	2	80,687.900
Ba II	ns 2S J=1/2	6-30†	18	16	14/14	2.3-26.4	+3.5984	0.0191	2	80,687.900
Ba III	5p5.(2P<3/2>).nd 2[1/2] J=0	5-7	3	1	1/1	2.5-4.7	+2.3509	0.0972	3	289,100.000
Ba III	5p5.(2P<3/2>).nd 2[1/2] J=1	5-7	3	1	1/1	2.5-4.8	+2.3360	0.0975	3	289,100.000
Ba III	5p5.(2P<3/2>).nd 2[3/2] J=1	5-7	3	1	1/1	2.7-4.9	+2.1891	0.0829	3	289,100.000
Ba III	5p5.(2P<3/2>).nd 2[3/2] J=2	5-7	3	1	1/1	2.6-4.8	+2.3061	0.0981	3	289,100.000
Ba III	5p5.(2P<3/2>).nd 2[5/2] J=2	5-7	3	1	1/1	2.6-4.8	+2.2821	0.0849	3	289,100.000
Ba III	5p5.(2P<3/2>).nd 2[5/2] J=3	5-7	3	1	1/1	2.6-4.8	+2.2601	0.0738	3	289,100.000
Ba III	5p5.(2P<3/2>).nd 2[7/2] J=3	5-7	3	1	1/1	2.6-4.8	+2.3041	0.0846	3	289,100.000
Ba III	5p5.(2P<3/2>).nd 2[7/2] J=4	5-7	3	1	1/1	2.6-4.8	+2.3113	0.0890	3	289,100.000
Ba III	5p5.(2P<3/2>).ns 2[3/2] J=1	6-8	3	1	1/1	2.7-4.8	+3.2585	0.0242	3	289,100.000
Ba III	5p5.(2P<3/2>).ns 2[3/2] J=2	6-8	3	1	1/1	2.7-4.7	+3.2795	0.0267	3	289,100.000
Ba III	nd 2[3/2]* J=2	7-16	10	8	8/8	4.8-13.8	+2.1786	0.0136	3	306,650.000
Ba III	ns 2[1/2]* J=1	8-17	10	8	untested	4.8-13.8	+3.2300	0.0154	3	306,650.000
Ba III	nd 2[3/2]* J=2	8-22†	14	12	untested	5.9-19.9	+2.1273	0.0227	3	289,100.000
Ba III	ns 2[3/2]* J=1	9-23	15	13	untested	5.8-19.9	+3.2018	0.0273	3	289,100.000
Ba III *	ng 2[11/2]* J=5	5-6	2	1	untested	5.0-6.0	+0.0394	0.0028	3	289,100.000
Ba III *	ng 2[11/2]* J=6	5-6	2	1	untested	5.0-6.0	+0.0397	0.0029	3	289,100.000
Ba III *	ng 2[5/2]* J=2	5-6	2	1	untested	5.0-6.0	+0.0428	0.0002	3	289,100.000
Ba III *	ng 2[5/2]* J=3	5-6	2	1	untested	5.0-6.0	+0.0422	0.0000	3	289,100.000
Ba III *	ng 2[7/2]* J=3	5-6	2	1	untested	5.0-6.0	+0.0208	0.0006	3	289,100.000
Ba III *	ng 2[7/2]* J=4	5-6	2	1	untested	5.0-6.0	+0.0212	0.0007	3	289,100.000
Ba III *	ng 2[9/2]* J=4	5-6	2	1	untested	5.0-6.0	+0.0174	0.0030	3	289,100.000
Ba III *	ng 2[9/2]* J=5	5-6	2	1	untested	5.0-6.0	+0.0172	0.0029	3	289,100.000
Be I	2snp ¹ P°	3-13	11	9	9/9	2.7-12.6	+0.3590	0.1025	1	75,192.500
Be I	2snd ³ D	3-12	10	8	8/8	2.9-11.9	+0.1109	0.0129	1	75,192.500
Be I	2snd ¹ D	3-12	10	8	8/8	3.2-12.1	-0.1152	0.0970	1	75,192.500
Be I	2sns ¹ S	3-11	9	7	7/7	2.3-10.3	+0.6774	0.0136	1	75,192.500
Be I	2sns ³ S	3-8	6	4	4/4	2.2-7.2	+0.7910	0.0429	1	75,192.500
Be I	2snp ³ P°	3-8	6	4	4/4	2.6-7.6	+0.3773	0.0374	1	75,192.500
Be I	2snf ³ F°	4-7	4	2	2/2	4.0-7.0	+0.0305	0.0063	1	75,192.500
Be II	ns ² S	2-10	9	7	7/7	1.7-9.7	+0.2623	0.0112	2	146,882.860
Be II	np ² P°	2-10	9	7	7/7	2.0-10.0	+0.0491	0.0039	2	146,882.860
Be II	nd ² D	3-9	7	5	5/5	3.0-9.0	+0.0021	0.0004	2	146,882.860
Be II	nf ² F°	4-10	7	5	5/5	4.0-10.0	+0.0001	0.0001	2	146,882.860
Be II	ng ² G	5-10	6	4	4/4	5.0-10.0	-0.0001	0.0001	2	146,882.860
Be II	nh ² H°	6-10	5	3	3/3	6.0-10.0	-0.0002	0.0001	2	146,882.860
Be III	1s.nd 1D J=2	3-10	8	6	4/6	3.0-10.0	+0.0008	0.0002	3	1,241,256.601
Be III	1s.nd 3D J=2	3-10	8	6	3/6	3.0-10.0	+0.0023	0.0002	3	1,241,256.601
Be III	1s.nf 1F* J=3	4-9	6	4	2/4	4.0-9.0	+0.0002	0.0001	3	1,241,256.601
Be III	1s.ng 1G J=4	5-9	5	3	1/3	5.0-9.0	+0.0002	0.0001	3	1,241,256.601
Be III	1s.nh 1H* J=5	6-9	4	2	2/2	6.0-9.0	+0.0000	0.0000	3	1,241,256.601
Be III *	1s.ni 1I J=6	7-8	2	1	no-triple	7.0-8.0	+0.0000	0.0000	3	1,241,256.601
Be III	1s.np 3P* J=1	2-10	9	7	5/7	2.0-10.0	+0.0429	0.0001	3	1,241,256.601
Be III	1s.ns 1S J=0	2-10	9	7	4/7	1.9-9.9	+0.0508	0.0003	3	1,241,256.601
Be III	1s.ns 3S J=1	2-10	9	7	5/7	1.9-9.9	+0.1302	0.0028	3	1,241,256.601
Be IV	nd 2D J=3/2	3-10	8	6	6/6	3.0-10.0	+0.0002	0.0000	4	1,756,018.810
Be IV	nf 2F* J=5/2	4-10	7	5	5/5	4.0-10.0	+0.0001	0.0000	4	1,756,018.810
Be IV	ng 2G J=7/2	5-10	6	4	4/4	5.0-10.0	+0.0001	0.0000	4	1,756,018.810
Be IV	nh 2H* J=9/2	6-10	5	3	3/3	6.0-10.0	+0.0000	0.0000	4	1,756,018.810
Be IV	ni 2I J=11/2	7-10	4	2	2/2	7.0-10.0	+0.0000	0.0000	4	1,756,018.810
Be IV	nk 2K* J=13/2	8-10	3	1	1/1	8.0-10.0	+0.0000	0.0000	4	1,756,018.810
Be IV	np 2P* J=1/2	3-10	8	6	6/6	3.0-10.0	+0.0004	0.0000	4	1,756,018.810
Be IV	ns 2S J=1/2	3-10	8	6	6/6	3.0-10.0	+0.0004	0.0000	4	1,756,018.810
Bi I	(³ Po)ns 2[0] J=1/2	7-11	5	3	3/3	2.0-6.1	+4.9036	0.0649	1	58,761.650
Bi I	(³ Po)nd 2[2] J=3/2	6-10	5	3	3/3	2.7-6.7	+3.2646	0.0470	1	58,761.650
Bi I	(³ Po)nd 2[2] J=5/2	6-9	4	2	2/2	2.8-5.8	+3.2071	0.0272	1	58,761.650
Bi I	(³ Po)np 2[1]° J=1/2	7-9	3	1	1/1	2.5-4.6	+4.4712	0.0588	1	58,761.650
Bi II	6s2.6p.nd (1/2,3/2)* J=1	6-14	9	7	2/7	2.8-11.0	+2.9846	0.0626	2	134,720.000
Bi II	6s2.6p.nd (1/2,3/2)* J=2	6-10†	4	2	no-triple	2.8-6.9	+3.1079	0.0545	2	134,720.000
Bi II	6s2.6p.nd (1/2,5/2)* J=2	6-8	3	1	0/1	2.9-5.0	+3.0001	0.1072	2	134,720.000
Bi II	6s2.6p.nf (1/2,5/2) J=3	5-8†	3	1	no-triple	3.9-6.8	+1.1755	0.0437	2	134,720.000
Bi II	6s2.6p.nf (1/2,7/2) J=4	5-8†	3	1	no-triple	3.9-6.8	+1.1468	0.0343	2	134,720.000
Bi II	6s2.6p.ns (1/2,1/2)* J=0	7-9	3	1	1/1	2.6-4.7	+4.3705	0.0333	2	134,720.000
Bi II	6s2.6p.ns (1/2,1/2)* J=1	7-9	3	1	1/1	2.6-4.7	+4.3684	0.0313	2	134,720.000
Bi II *	6s2.6p.nf (1/2,7/2) J=3	5-6	2	1	no-triple	3.9-4.9	+1.1419	0.0039	2	134,720.000
Bi II *	6s2.6p.np (1/2,1/2) J=0	7-8	2	1	no-triple	3.0-4.1	+3.9567	0.0079	2	134,720.000
Bi III	6s2.nd 2D J=3/2	6-8	3	1	1/1	3.0-5.2	+2.8926	0.0795	3	206,242.000
Bi III	6s2.nd 2D J=5/2	6-8	3	1	1/1	3.1-5.2	+2.8456	0.0496	3	206,242.000
Bi III	6s2.ng 2G J=7/2	5-8	4	2	2/2	5.0-8.0	+0.0378	0.0043	3	206,242.000
Bi III	6s2.ng 2G J=9/2	5-8	4	2	2/2	5.0-8.0	+0.0383	0.0046	3	206,242.000
Bi III	6s2.np 2P* J=1/2	6-8	3	1	1/1	2.2-4.4	+3.7096	0.0732	3	206,242.000
Bi III	6s2.np 2P* J=3/2	6-8	3	1	1/1	2.3-4.5	+3.6044	0.0630	3	206,242.000
Bi III	6s2.ns 2S J=1/2	7-9	3	1	1/1	3.0-5.0	+3.9869	0.0234	3	206,242.000
Bi III *	6s2.nh 2H* J=11/2	6-7	2	1	no-triple	6.0-7.0	+0.0103	0.0005	3	206,242.000
Bi III *	6s2.nh 2H* J=9/2	6-7	2	1	no-triple	6.0-7.0	+0.0103	0.0005	3	206,242.000
C I *	nd 1F* J=3	3-4	2	1	no-triple	3.0-4.0	+0.0320	0.0059	1	90,937.500
C I *	nd 3D* J=1	3-4	2	1	no-triple	2.9-3.9	+0.0638	0.0097	1	90,937.500
C I *	np 3D J=1	3-4	2	1	no-triple	2.3-3.3	+0.7201	0.0073	1	90,937.500
C I *	np 3P J=0	3-4	2	1	no-triple	2.4-3.4	+0.6284	0.0046	1	90,937.500
C I *	np 3S J=1	3-4	2	1	no-triple	2.3-3.3	+0.6641	0.0048	1	90,937.500
C I	ns 1P* J=1	3-5	3	1	0/1	1.9-3.9	+1.0532	0.0037	1	90,937.500
C I	ns 3P* J=0	3-5	3	1	0/1	1.9-3.9	+1.0925	0.0127	1	90,937.500
C II	2s ² ns ² S	3-8	6	4	4/4	2.3-7.3	+0.6624	0.0063	2	196,664.700
C II	2s ² np ² P°	3-7	5	3	3/3	2.6-6.7	+0.3928	0.1344	2	196,664.700
C II	2s ² nd ² D	3-7	5	3	3/3	2.9-6.9	+0.0924	0.0645	2	196,664.700

The Method 1.6

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
C II	2s ² nf 2P°	4-7	4	2	2/2	4.0-7.0	+0.0220	0.0069	2	196,664.700
C II	2s ² ng 2G	5-7	3	1	1/1	5.0-7.0	+0.0054	0.0011	2	196,664.700
C II	2s2(1S)np 2P° J=1/2	2-4	3	1	0/1	1.5-3.6	+0.4403	0.0469	2	196,664.700
C II	2s2(1S)np 2P° J=3/2	2-4	3	1	0/1	1.5-3.6	+0.4401	0.0469	2	196,664.700
C II *	2s2(1S)nd 2D J=3/2	3-4	2	1	no-triple	2.9-3.9	+0.0740	0.0044	2	196,664.700
C II *	2s2(1S)nd 2D J=5/2	3-4	2	1	no-triple	2.9-3.9	+0.0740	0.0044	2	196,664.700
C II *	2s2(1S)ns 2S J=1/2	3-4	2	1	no-triple	2.3-3.3	+0.6615	0.0020	2	196,664.700
C III	nd 1D J=2	3-7	5	3	1/3	3.0-7.0	+0.0067	0.0036	3	386,241.000
C III	nd 3D J=1	3-9	7	5	3/5	2.9-8.9	+0.0816	0.0069	3	386,241.000
C III	nf 3F* J=2	4-7	4	2	1/2	3.9-7.0	+0.0127	0.0398	3	386,241.000
C III	ng 3G J=3	5-7	3	1	1/1	5.0-7.0	+0.0126	0.0021	3	386,241.000
C III	np 1P* J=1	3-8	6	4	1/4	2.8-7.8	+0.1806	0.0527	3	386,241.000
C III	np 3P* J=0	3-6	4	2	1/2	2.8-5.8	+0.1856	0.0217	3	386,241.000
C III	ns 1S J=0	3-5	3	1	1/1	2.7-4.5	+0.3819	0.0499	3	386,241.000
C III	ns 3S J=1	3-7	5	3	2/3	2.6-6.6	+0.3993	0.0137	3	386,241.000
C V	1s.nd 1D J=2	3-7	5	3	3/3	3.0-7.0	+0.0108	0.0077	5	3,162,788.800
C V	1s.nd 3D J=1	3-7	5	3	3/3	3.0-7.0	+0.0122	0.0078	5	3,162,788.800
C V	1s.nf 1F* J=3	4-6	3	1	1/1	4.0-6.0	+0.0092	0.0042	5	3,162,788.800
C V	1s.nf 3F* J=3	4-6	3	1	1/1	4.0-6.0	+0.0092	0.0041	5	3,162,788.800
C V	1s.ng 1G J=4	5-7	3	1	1/1	5.0-7.0	+0.0152	0.0059	5	3,162,788.800
C V	1s.ng 3G J=4	5-7	3	1	1/1	5.0-7.0	+0.0153	0.0059	5	3,162,788.800
C V *	1s.nh 1H* J=5	6-7	2	1	no-triple	6.0-7.0	+0.0186	0.0042	5	3,162,788.800
C V	1s.np 1P* J=1	2-7	6	4	4/4	2.0-7.0	+0.0002	0.0084	5	3,162,788.800
C V	1s.np 3P* J=0	2-7	6	4	4/4	2.0-6.9	+0.0384	0.0075	5	3,162,788.800
C V	1s.ns 1S J=0	2-7	6	4	4/4	2.0-6.9	+0.0399	0.0079	5	3,162,788.800
C V	1s.ns 3S J=1	2-7	6	4	3/4	1.9-6.9	+0.0927	0.0063	5	3,162,788.800
Ca I	nd 1D J=2	4-9	6	4	untested	3.0-8.6	+0.9084	0.2360	1	49,305.950
Ca I	nd 3D J=1	4-9	6	4	untested	3.1-8.4	+0.8917	0.1405	1	49,305.950
Ca I	nf 1F* J=3	4-8	5	3	untested	4.0-7.9	+0.0654	0.0190	1	49,305.950
Ca I	nf 3F* J=2	4-9	6	4	untested	3.9-8.9	+0.0893	0.0059	1	49,305.950
Ca I	np 1P* J=1	5-10	6	4	untested	3.0-8.2	+1.8902	0.1793	1	49,305.950
Ca I	ns 1S J=0	5-11	7	5	untested	2.6-8.7	+2.3548	0.0350	1	49,305.950
Ca I	ns 3S J=1	5-11	7	5	untested	2.5-8.6	+2.4639	0.0227	1	49,305.950
Ca II	ns 2S J=1/2	4-6	3	1	1/1	2.1-4.2	+1.8338	0.0183	2	95,751.880
Ca II	nd 2D J=3/2	3-16	14	12	untested	2.3-15.4	+0.6341	0.0154	2	95,751.870
Ca II	nf 2F* J=5/2	4-10	7	5	untested	4.0-10.0	+0.0248	0.0029	2	95,751.870
Ca II	ng 2G J=7/2	5-9	5	3	untested	5.0-9.0	+0.0048	0.0004	2	95,751.870
Ca II *	nh 2H* J=9/2	8-10†	2	1	untested	8.0-10.0	+0.0016	0.0000	2	95,751.870
Ca II	np 2P* J=1/2	4-6	3	1	untested	2.5-4.5	+1.4775	0.0207	2	95,751.870
Ca II	ns 2S J=1/2	4-10	7	5	untested	2.1-8.2	+1.8192	0.0175	2	95,751.870
Ca IX	nd 1D J=2	4-6	3	1	0/1	3.9-5.8	+0.1572	0.0090	9	1,520,640.000
Ca IX *	nf 1F* J=3	4-5	2	1	no-triple	4.0-5.0	+0.0124	0.0050	9	1,520,640.000
Ca IX	np 1P* J=1	4-6	3	1	0/1	3.6-5.6	+0.4121	0.0041	9	1,520,640.000
Ca IX *	ns 3S J=1	4-5	2	1	no-triple	3.4-4.4	+0.5725	0.0078	9	1,520,640.000
Cd I	nd 1D J=2	5-15†	10	8	5/6	2.9-12.8	+2.1886	0.0288	1	72,540.050
Cd I	nd 3D J=1	5-11	7	5	5/5	2.9-8.9	+2.0898	0.0049	1	72,540.050
Cd I	nf 3F* J=3	4-10	7	5	5/5	4.0-10.0	+0.0346	0.0034	1	72,540.050
Cd I	np 1P* J=1	6-12	7	5	3/5	2.9-8.9	+3.0515	0.0005	1	72,540.050
Cd I	np 3P* J=0	6-10	5	3	3/3	2.8-6.8	+3.1814	0.0184	1	72,540.050
Cd I	ns 1S J=0	6-15	10	8	5/8	2.4-11.4	+3.5926	0.0131	1	72,540.050
Cd I	ns 3S J=1	6-16	11	9	7/9	2.3-12.3	+3.6677	0.0188	1	72,540.050
Cd II	nd 2D J=3/2	5-13	9	7	7/7	3.1-11.1	+1.8942	0.0159	2	136,374.740
Cd II	nf 2F* J=5/2	4-9	6	4	4/4	4.0-8.9	+0.0489	0.0092	2	136,374.740
Cd II	ng 2G J=7/2	5-11	7	5	4/5	5.0-11.0	+0.0076	0.0004	2	136,374.740
Cd II	np 2P* J=1/2	5-12†	7	5	2/3	2.2-9.3	+2.7444	0.0319	2	136,374.740
Cd II	ns 2S J=1/2	5-13	9	7	7/7	1.8-9.9	+3.1175	0.0330	2	136,374.740
Fe I	nd 4D J=7/2	3-6	4	2	1/2	3.0-6.0	+0.0323	0.0008	1	140,518.700
Fe I	nd 4F J=9/2	3-6	4	2	1/2	3.0-6.0	+0.0149	0.0012	1	140,518.700
Fe I *	np 4D* J=7/2	3-4	2	1	no-triple	2.2-3.2	+0.8308	0.0097	1	140,518.700
Fe I	ns 2P J=3/2	3-6	4	2	2/2	1.8-4.9	+1.1857	0.0604	1	140,518.700
Fe I	ns 4P J=5/2	3-8	6	4	4/4	1.7-6.7	+1.2800	0.0117	1	140,518.700
Fe XV *	nd 3D J=1	4-5	2	1	no-triple	3.9-4.9	+0.1278	0.0017	15	3,679,500.000
Fe XV	nf 1F* J=3	4-6	3	1	1/1	4.0-6.5	-0.2367	0.0230	15	3,679,500.000
Fe XV *	nf 3F* J=2	4-5	2	1	no-triple	4.0-5.0	+0.0371	0.0016	15	3,679,500.000
Fe XV *	np 1P* J=1	4-5	2	1	no-triple	3.7-4.7	+0.2872	0.0017	15	3,679,500.000
Fe XVI	nd 2D J=3/2	3-8	6	4	4/4	2.9-7.9	+0.0750	0.0031	16	3,946,570.000
Fe XVI	nf 2F* J=5/2	4-8	5	3	3/3	4.0-8.0	+0.0116	0.0036	16	3,946,570.000
Fe XVI	np 2P* J=1/2	3-6	4	2	2/2	2.8-5.8	+0.2233	0.0059	16	3,946,570.000
Fe XVI	ns 2S J=1/2	3-7	5	3	3/3	2.7-6.7	+0.3174	0.0152	16	3,946,570.000
Ga I	4s ² ns 2S	5-28	24	22	22/22	2.2-25.2	+2.8013	0.0535	1	48,387.634
Ga I	4s ² nd 2D	4-27	23	21	21/21	2.8-25.7	+1.2926	0.1920	1	48,387.634
Ga I	4s ² np 2P° high	41-5	15	13	13/13	38.8-52.7	+2.2107	0.1209	1	48,387.634
Ga I	4s ² nf 2F°	4-8	5	3	3/3	4.0-8.0	+0.0252	0.0057	1	48,387.634
Ga I	4s ² np 2P° low	4-7	4	2	2/2	1.5-4.7	+2.3450	0.2026	1	48,387.634
Ga II	nd 1D J=2	4-7	4	2	2/2	3.3-6.1	+0.8475	0.1127	2	165,465.800
Ga II	nd 3D J=1	4-7	4	2	2/2	2.9-5.9	+1.0630	0.0133	2	165,465.800
Ga II	nf 1F* J=3	4-7	4	2	2/2	4.0-6.9	+0.0598	0.0070	2	165,465.800
Ga II	nf 3F* J=2	4-7	4	2	2/2	3.9-6.9	+0.0617	0.0080	2	165,465.800
Ga II	ng (1/2, 7/2) J=3	5-8	4	2	2/2	5.0-8.0	+0.0146	0.0012	2	165,465.800
Ga II	ng (1/2, 9/2) J=5	5-8	4	2	2/2	5.0-8.0	+0.0142	0.0010	2	165,465.800
Ga II	ns 1S J=0	5-8	4	2	1/2	2.7-5.8	+2.2610	0.0138	2	165,465.800
Ga II	ns 3S J=1	5-8	4	2	2/2	2.6-5.7	+2.3228	0.0169	2	165,465.800
Ge III	nd 1D J=2	4-6	3	1	1/1	3.2-5.1	+0.8820	0.0572	3	274,693.000
Ge III	nd 3D J=1	4-6	3	1	1/1	3.0-5.0	+1.0025	0.0189	3	274,693.000
Ge III *	ng 3G J=3	5-6	2	1	no-triple	5.0-6.0	+0.0181	0.0010	3	274,693.000
Ge III	ns 1S J=0	5-7	3	1	1/1	3.0-5.0	+2.0116	0.0057	3	274,693.000
Ge III	ns 3S J=1	5-8	4	2	2/2	2.9-6.0	+2.0576	0.0162	3	274,693.000
He I	1sns 2S	2-35	25	23	23/23	1.7-34.7	+0.2965	0.0164	1	198,310.666
He I	1sns 1S	2-35	25	23	23/23	1.9-34.9	+0.1392	0.0118	1	198,310.666
He I	1snp 1P°	2-35	25	23	23/23	2.0-35.0	-0.0133	0.0050	1	198,310.666
He I	1snd 1D	3-35	24	22	22/22	3.0-35.0	+0.0007	0.0019	1	198,310.666

The Method 1.6

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
He I	1snf ¹ F°	4-35	23	21	21/21	4.0-35.0	-0.0010	0.0020	1	198,310.666
He I	1snd ³ D	3-35	22	20	20/20	3.0-35.0	+0.0014	0.0018	1	198,310.666
He I	1sng ¹ G	5-35	22	20	20/20	5.0-35.0	-0.0014	0.0020	1	198,310.666
He I	1snh ¹ H°	6-35	21	19	19/19	6.0-35.0	-0.0015	0.0020	1	198,310.666
He I	1sni ¹ I	7-35	20	18	18/18	7.0-35.0	-0.0016	0.0019	1	198,310.666
He II	ns	1-10	10	8	8/8	1.0-10.0	+7.6e-05	2e-05	2	438,908.871
He II	np	2-10	9	7	7/7	2.0-10.0	+5.1e-05	8e-06	2	438,908.871
He II	nd	3-10	8	6	6/6	3.0-10.0	+2.5e-05	4e-06	2	438,908.871
He II	nf	4-10	7	5	5/5	4.0-10.0	+1.4e-05	2e-06	2	438,908.871
He II	ng	5-10	6	4	4/4	5.0-10.0	+8.6e-06	1e-06	2	438,908.871
He II	nh	6-9	4	2	2/2	6.0-9.0	+4.5e-06	4e-07	2	438,908.871
He II	ni	7-9	3	1	1/1	7.0-9.0	+1.6e-06	4e-07	2	438,908.871

He II's seven rows print δ in exponent form because the series lies below the table's four decimals. They are the corrected series (register 1758): term values against the limit 438,908.871 cm⁻¹ with the reduced-mass Rydberg $R_{He} = R_{\infty}/(1 + m_e/M_{He})$, each level taken as the J-centroid of its fine-structure pair, from the NIST ASD levels; the earlier rows, -0.0003 to -0.0005 and rising, were the same levels against R_{∞} , which is the uncorrected series §24.4 replaced.

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Hg II	nd 2D J=3/2	6-11	6	4	4/4	3.1-8.1	+2.8758	0.0227	2	151,269.200
Hg II	nf 2F* J=7/2	5-8	4	2	2/2	4.0-6.9	+1.0622	0.0083	2	151,269.200
Hg II	ng 2G J=9/2	5-9	5	3	3/3	5.0-9.0	+0.0053	0.0028	2	151,269.200
Hg II	np 2P* J=1/2	6-8	3	1	1/1	2.1-4.2	+3.8196	0.0624	2	151,269.200
Hg II	ns 2S J=1/2	6-11	6	4	4/4	1.7-6.8	+4.1850	0.0515	2	151,269.200
K I	nd 2D J=5/2	3-13	11	9	9/9	2.9-12.7	+0.2460	0.0380	1	35,010.600
K I	nf 2F* J=5/2	4-12	9	7	7/7	4.0-12.0	+0.0112	0.0030	1	35,010.600
K I	np 2P* J=1/2	4-15	12	10	9/10	2.2-13.3	+1.7268	0.0133	1	35,010.600
K I	ns 2S J=1/2	4-18	15	13	12/13	1.8-15.8	+2.1912	0.0111	1	35,010.600
K II	3p ⁵ nf 2[3/2] J=1	4-6	3	1	1/1	4.0-6.0	+0.0426	0.0069	2	255,072.800
K II	3p ⁵ nf 2[9/2] J=5	4-6	3	1	1/1	4.0-6.0	+0.0336	0.0071	2	255,072.800
K II	3p ⁵ nf 2[5/2] J=3	4-6	3	1	1/1	4.0-6.0	+0.0242	0.0052	2	255,072.800
K II	3p ⁵ nf 2[7/2] J=4	4-6	3	1	1/1	4.0-6.0	+0.0143	0.0059	2	255,072.800
Li I	np 2P°	2-42	41	39	39/39	2.0-41.8	+0.1477	0.6779	1	43,487.114
Li I	ns 2S	2-11	10	8	8/8	1.6-10.6	+0.4014	0.0180	1	43,487.114
Li I	nd 2D	3-12	10	8	8/8	3.0-12.0	+0.0031	0.0129	1	43,487.114
Li I	np 2P° J=1/2	2-4	3	1	1/1	2.0-4.0	+0.0436	0.0021	1	43,487.150
Li I	np 2P° J=3/2	2-4	3	1	1/1	2.0-4.0	+0.0436	0.0021	1	43,487.150
Li I *	nd 2D J=3/2	3-4	2	1	no-triple	3.0-4.0	+0.0016	0.0001	1	43,487.150
Li I *	nd 2D J=5/2	3-4	2	1	no-triple	3.0-4.0	+0.0016	0.0001	1	43,487.150
Li I *	ns 2S J=1/2	2-3	2	1	no-triple	1.6-2.6	+0.4077	0.0038	1	43,487.150
Li II	1s.nd 1D J=2	3-10	8	6	6/6	3.0-10.0	+0.0011	0.0001	2	610,078.400
Li II	1s.nd 3D J=2	3-10	8	6	5/6	3.0-10.0	+0.0027	0.0003	2	610,078.400
Li II	1s.nf 1F* J=3	4-10	7	5	5/5	4.0-10.0	+0.0000	0.0002	2	610,078.400
Li II	1s.nf 3F* J=3	4-10	7	5	4/5	4.0-10.0	+0.0006	0.0002	2	610,078.400
Li II	1s.ng 1G J=4	5-9	5	3	3/3	5.0-9.0	+0.0000	0.0000	2	610,078.400
Li II	1s.nh 1H* J=5	6-9	4	2	2/2	6.0-9.0	-0.0000	0.0000	2	610,078.400
Li II	1s.np 1P* J=1	2-14	13	11	7/11	2.0-14.0	-0.0144	0.0014	2	610,078.400
Li II	1s.np 3P* J=1	2-10	9	7	7/7	1.9-9.9	+0.0543	0.0004	2	610,078.400
Li II	1s.ns 1S J=0	2-10	9	7	5/7	1.9-9.9	+0.0744	0.0010	2	610,078.400
Li II	1s.ns 3S J=1	2-10	9	7	7/7	1.8-9.8	+0.1814	0.0035	2	610,078.400
Li II *	1s.ni 1I J=6	7-8	2	1	no-triple	7.0-8.0	-0.0001	0.0000	2	610,078.400
Li II *	1s.np 3P* J=0	2-3	2	1	no-triple	1.9-2.9	+0.0537	0.0004	2	610,078.400
Li II *	1s.np 3P* J=2	2-3	2	1	no-triple	1.9-2.9	+0.0537	0.0004	2	610,078.400
Li III	nd 2D J=3/2	3-9	7	5	4/5	3.0-9.0	+0.0003	0.0002	3	987,662.290
Li III	nd 2D J=5/2	3-9	7	5	4/5	3.0-9.0	+0.0002	0.0002	3	987,662.290
Li III	nf 2F* J=5/2	4-9	6	4	4/4	4.0-9.0	+0.0003	0.0002	3	987,662.290
Li III	nf 2F* J=7/2	4-9	6	4	4/4	4.0-9.0	+0.0002	0.0002	3	987,662.290
Li III	ng 2G J=7/2	5-9	5	3	3/3	5.0-9.0	+0.0003	0.0001	3	987,662.290
Li III	ng 2G J=9/2	5-9	5	3	3/3	5.0-9.0	+0.0003	0.0001	3	987,662.290
Li III	nh 2H* J=11/2	6-9	4	2	2/2	6.0-9.0	+0.0003	0.0001	3	987,662.290
Li III	nh 2H* J=9/2	6-9	4	2	2/2	6.0-9.0	+0.0003	0.0001	3	987,662.290
Li III	ni 2I J=11/2	7-9	3	1	1/1	7.0-9.0	+0.0004	0.0001	3	987,662.290
Li III	ni 2I J=13/2	7-9	3	1	1/1	7.0-9.0	+0.0004	0.0001	3	987,662.290
Li III	np 2P* J=1/2	2-10	9	7	5/7	2.0-10.0	+0.0004	0.0002	3	987,662.290
Li III	np 2P* J=3/2	2-10	9	7	5/7	2.0-10.0	+0.0003	0.0002	3	987,662.290
Li III	ns 2S J=1/2	1-9	9	7	5/7	1.0-9.0	+0.0003	0.0002	3	987,662.290
Li III *	nk 2K* J=13/2	8-9	2	1	no-triple	8.0-9.0	+0.0004	0.0001	3	987,662.290
Li III *	nk 2K* J=15/2	8-9	2	1	no-triple	8.0-9.0	+0.0004	0.0001	3	987,662.290
Mg I	3s.nd 1D J=2	3-5	3	1	1/1	2.7-4.5	+0.4034	0.0647	1	61,671.050
Mg I	3s.nd 3D J=1	3-5	3	1	1/1	2.8-4.8	+0.1701	0.0009	1	61,671.050
Mg I	3s.nd 3D J=2	3-5	3	1	1/1	2.8-4.8	+0.1701	0.0008	1	61,671.050
Mg I	3s.nd 3D J=3	3-5	3	1	1/1	2.8-4.8	+0.1701	0.0008	1	61,671.050
Mg I	3snp 1P° J=1	3-6	4	2	2/2	2.0-5.0	+1.0135	0.0265	1	61,671.050
Mg I	3snp 3P° J=0	3-6	4	2	2/2	1.7-4.9	+1.2056	0.0789	1	61,671.050
Mg I	3snp 3P° J=1	3-6	4	2	2/2	1.7-4.9	+1.2052	0.0789	1	61,671.050
Mg I	3snp 3P° J=2	3-6	4	2	2/2	1.7-4.9	+1.2045	0.0788	1	61,671.050
Mg I	3sns 1S J=0	4-6	3	1	1/1	2.5-4.5	+1.5331	0.0067	1	61,671.050
Mg I	3sns 3S J=1	4-6	3	1	1/1	2.3-4.4	+1.6603	0.0180	1	61,671.050
Mg I *	3snf 1F° J=3	4-5	2	1	no-triple	4.0-5.0	+0.0413	0.0022	1	61,671.050

The Method 1.6

species	series	n	levels	interior	bracket	n* range	6	$\sigma(6)$	fits	limit cm ⁻¹
Mg I *	3snf 3F° J=2	4-5	2	1	no-triple	4.0-5.0	+0.0413	0.0022	1	61,671.050
Mg I *	3snf 3F° J=3	4-5	2	1	no-triple	4.0-5.0	+0.0413	0.0022	1	61,671.050
Mg I *	3snf 3F° J=4	4-5	2	1	no-triple	4.0-5.0	+0.0413	0.0022	1	61,671.050
Mg II	ns 2S	3-10	8	6	6/6	1.9-8.9	+1.0749	0.0289	2	121,267.640
Mg II	nd 2D	3-10	8	6	6/6	3.0-10.0	+0.0411	0.0145	2	121,267.640
Mg II	np 2P°	3-9	7	5	5/5	2.3-8.3	+0.7077	0.0361	2	121,267.640
Mg II	nf 2F°	4-10	7	5	5/5	4.0-10.0	+0.0030	0.0009	2	121,267.640
Mg II	ng 2G	5-10	6	4	4/4	5.0-10.0	+0.0007	0.0001	2	121,267.640
Mg II	nh 2H°	6-9	4	2	2/2	6.0-9.0	+0.0002	0.0000	2	121,267.640
Mg II	ni 2I	7-9	3	1	1/1	7.0-9.0	+0.0000	0.0000	2	121,267.640
Mg II	np 2P° J=1/2	3-5	3	1	1/1	2.3-4.3	+0.7187	0.0123	2	121,267.640
Mg II	np 2P° J=3/2	3-5	3	1	1/1	2.3-4.3	+0.7174	0.0123	2	121,267.640
Mg II	ns 2S J=1/2	3-6	4	2	2/2	1.9-4.9	+1.0806	0.0102	2	121,267.640
Mg II *	nd 2D J=3/2	3-4	2	1	no-triple	3.0-4.0	+0.0340	0.0036	2	121,267.640
Mg II *	nd 2D J=5/2	3-4	2	1	no-triple	3.0-4.0	+0.0340	0.0036	2	121,267.640
Mg III *	nd 2[1/2]* J=1	4-5	2	1	no-triple	3.9-4.9	+0.0934	0.0014	3	646,402.000
Mg III	nd 2[3/2]* J=1	4-9	6	4	1/4	4.0-9.0	+0.0388	0.0017	3	646,402.000
Mg III *	nf 2[3/2] J=1	4-5	2	1	no-triple	4.0-5.0	+0.0111	0.0005	3	646,402.000
Mg III *	nf 2[3/2] J=2	4-5	2	1	no-triple	4.0-5.0	+0.0108	0.0004	3	646,402.000
Mg III *	nf 2[5/2] J=3	4-5	2	1	no-triple	4.0-5.0	+0.0039	0.0003	3	646,402.000
Mg III *	nf 2[7/2] J=3	4-5	2	1	no-triple	4.0-5.0	+0.0000	0.0004	3	646,402.000
Mg III *	nf 2[7/2] J=4	4-5	2	1	no-triple	4.0-5.0	-0.0000	0.0004	3	646,402.000
Mg III *	nf 2[9/2] J=4	4-5	2	1	no-triple	4.0-5.0	+0.0073	0.0004	3	646,402.000
Mg III *	nf 2[9/2] J=5	4-5	2	1	no-triple	4.0-5.0	+0.0073	0.0004	3	646,402.000
Mg III	ns 2[3/2]* J=1	4-6	3	1	1/1	3.1-5.2	+0.8487	0.0050	3	646,402.000
Mg III *	ns 2[3/2]* J=2	4-5	2	1	no-triple	3.1-4.1	+0.8634	0.0030	3	646,402.000
N I	nd 2F J=5/2	3-5	3	1	1/1	3.0-5.0	+0.0273	0.0024	1	117,244.100
N I *	nd 4D J=1/2	3-4	2	1	no-triple	3.0-4.0	+0.0041	0.0041	1	117,244.100
N I	nd 4F J=3/2	3-5	3	1	0/1	3.0-4.9	+0.0514	0.0035	1	117,244.100
N I	np 2D* J=3/2	3-5	3	1	0/1	2.3-4.5	+0.6233	0.0955	1	117,244.100
N I	np 4D* J=1/2	3-5	3	1	1/1	2.2-4.2	+0.7714	0.0136	1	117,244.100
N I	np 4P* J=1/2	3-5	3	1	1/1	2.2-4.3	+0.7329	0.0169	1	117,244.100
N I	ns 2P J=1/2	3-6	4	2	2/2	1.9-4.9	+1.1040	0.0116	1	117,244.100
N I	ns 4P J=1/2	3-6	4	2	2/2	1.8-4.8	+1.1697	0.0190	1	117,244.100
N II	2p-ns 3P° J=0	3-6	4	2	2/2	2.2-5.2	+0.7747	0.0264	2	238,750.500
N II	2p-ns 3P° J=1	3-6	4	2	2/2	2.2-5.2	+0.7731	0.0260	2	238,750.500
N II	2p-ns 3P° J=2	3-6	4	2	2/2	2.2-5.3	+0.7633	0.0464	2	238,750.500
N II	2p-ns 1P° J=1	3-6	4	2	2/2	2.2-5.3	+0.7325	0.0824	2	238,750.500
N III	nd 2D J=3/2	3-12	10	8	5/8	2.9-11.9	+0.0737	0.0124	3	382,672.000
N III	nf 2F* J=5/2	4-7	4	2	2/2	4.0-7.0	+0.0281	0.0048	3	382,672.000
N III	ng 2G J=7/2	5-7	3	1	1/1	5.0-7.0	+0.0022	0.0009	3	382,672.000
N III *	nh 2H* J=9/2	6-7	2	1	no-triple	6.0-7.0	-0.0026	0.0009	3	382,672.000
N III	np 2P* J=1/2	3-5	3	1	0/1	2.7-4.7	+0.2917	0.0185	3	382,672.000
N III	ns 2S J=1/2	3-14	12	10	9/10	2.5-13.3	+0.5660	0.0740	3	382,672.000
Na I	ns	3-20	18	16	16/16	1.6-18.7	+1.3506	0.0250	1	41,449.451
Na I	np	3-20	18	16	16/16	2.1-19.1	+0.8584	0.0279	1	41,449.451
Na I	nd	3-20	18	16	16/16	3.0-20.0	+0.0141	0.0045	1	41,449.451
Na I	nf	4-20	17	15	15/15	4.0-20.0	+0.0014	0.0004	1	41,449.451
Na I	ng	5-20	16	14	14/14	5.0-20.0	+0.0003	0.0001	1	41,449.451
Na I	nh	6-10	5	3	3/3	6.0-10.0	+0.0000	0.0000	1	41,449.451
Na II	(2P° 3/2)ns 2[3/2]° J=2	3-6	4	2	2/2	1.9-5.0	+1.0385	0.0284	2	381,390.200
Na II	(2P° 3/2)ns 2[3/2]° J=1	3-6	4	2	2/2	1.9-5.0	+1.0268	0.0371	2	381,390.200
Na II	(2P° 3/2)nd 2[1/2]° J=0	3-5	3	1	1/1	2.9-4.9	+0.0671	0.0083	2	381,390.200
Na II	(2P° 3/2)nd 2[1/2]° J=1	3-5	3	1	1/1	2.9-4.9	+0.0641	0.0085	2	381,390.200
Na II	(2P° 3/2)nd 2[3/2]° J=2	3-5	3	1	1/1	2.9-4.9	+0.0556	0.0027	2	381,390.200
Na II	(2P° 3/2)nd 2[3/2]° J=1	3-5	3	1	1/1	3.0-5.0	+0.0268	0.0026	2	381,390.200
Ne I	2s22p5(2P3/2)nd 2[3/2] J=1	13-20	8	6	2/6	13.0-20.0	+0.0154	0.0024	1	173,929.750
Ne I	2s22p5(2P3/2)ns 2[3/2] J=1	12-20	9	7	4/7	10.7-18.7	+1.2923	0.0047	1	173,929.750
Ne I	nd 2[3/2]* J=1	11-20	10	8	untested	11.0-20.0	+0.0119	0.0043	1	174,710.090
Ne I	ns 2[1/2]* J=1	11-20	10	8	untested	9.7-18.7	+1.2974	0.0038	1	174,710.090
Ne I	np 1P* J=1	3-12	10	8	untested	2.2-11.1	+0.8409	0.0191	1	390,977.350
Ne I *	2p5(2P°3/2)nd 2[7/2]° J=4	3-4	2	1	no-triple	3.0-4.0	+0.0193	0.0014	1	173,929.750
Ne I *	2p5(2P°3/2)ns 2[3/2]° J=2	3-4	2	1	no-triple	1.7-2.7	+1.3329	0.0085	1	173,929.750
Ne II	nd 4D J=3/2	3-6	4	2	untested	2.9-5.9	+0.0700	0.0012	2	330,388.600
Ne II	nd 4D J=5/2	3-6	4	2	untested	2.9-5.9	+0.0773	0.0036	2	330,388.600
Ne II	nd 4D J=7/2	3-6	4	2	untested	2.9-5.9	+0.0813	0.0047	2	330,388.600
Ne II	nd 4F J=9/2	3-6	4	2	untested	3.0-5.9	+0.0547	0.0070	2	330,388.600
Ne II	ns 2P J=1/2	3-7	5	3	untested	2.0-6.3	+0.8636	0.0857	2	330,388.600
Ne II	ns 2P J=3/2	3-7	5	3	untested	2.0-6.0	+0.9480	0.0143	2	330,388.600
Ne II	ns 4P J=1/2	3-7	5	3	untested	2.0-6.2	+0.9132	0.0743	2	330,388.600
Ne II	ns 4P J=3/2	3-7	5	3	untested	2.0-6.2	+0.9261	0.0730	2	330,388.600
Ne II	ns 4P J=5/2	3-7	5	3	untested	2.0-6.0	+0.9906	0.0125	2	330,388.600
Ne II	nd 2S J=1/2	3-8	6	4	untested	3.0-7.9	+0.0701	0.0422	2	356,229.300
Ne II	ns 2D J=3/2	3-5	3	1	untested	2.0-4.0	+0.9850	0.0115	2	356,229.300
Ne II	ns 2D J=5/2	3-5	3	1	untested	2.0-4.0	+0.9850	0.0115	2	356,229.300
Ne II	np 2D* J=3/2	3-6	4	2	untested	2.4-5.5	+0.5651	0.0645	2	330,388.600
Ne II	np 2D* J=5/2	3-6	4	2	untested	2.4-5.4	+0.6294	0.0117	2	330,388.600
Ne II	np 2S* J=1/2	3-6	4	2	untested	2.4-5.4	+0.6010	0.0199	2	330,388.600
Ne II	np 4D* J=1/2	3-6	4	2	untested	2.3-5.5	+0.5916	0.0576	2	330,388.600
Ne II	np 4D* J=5/2	3-6	4	2	untested	2.3-5.5	+0.6055	0.0551	2	330,388.600

The Method 1.6

species	series	n	levels	interior	bracket	n* range	δ	$\sigma(\delta)$	fits	limit cm ⁻¹
Ne II	np 4D* J=7/2	3-6	4	2	untested	2.3-5.4	+0.6541	0.0133	2	330,388.600
Ne II	np 4P* J=1/2	3-6	4	2	untested	2.3-5.4	+0.6430	0.0585	2	330,388.600
Ne II	np 4P* J=3/2	3-6	4	2	untested	2.3-5.3	+0.6767	0.0235	2	330,388.600
Ne II	np 4P* J=5/2	3-6	4	2	untested	2.3-5.3	+0.6884	0.0174	2	330,388.600
Ne II	np 4S* J=3/2	3-6	4	2	untested	2.4-5.5	+0.5679	0.0424	2	330,388.600
Ne II	nf 2[2]* J=5/2	5-8	4	2	untested	5.0-8.0	+0.0039	0.0001	2	330,388.600
Ne II	nf 2[3]* J=7/2	5-8	4	2	untested	5.0-8.0	+0.0086	0.0001	2	330,388.600
Ne II	nf 2[4]* J=7/2	5-8	4	2	untested	5.0-8.0	+0.0096	0.0002	2	330,388.600
Ne II	nf 2[4]* J=9/2	5-8	4	2	untested	5.0-8.0	+0.0098	0.0002	2	330,388.600
Ne II	nf 2[5]* J=11/2	5-8	4	2	untested	5.0-8.0	+0.0025	0.0003	2	330,388.600
Ne II	ng 2[2] J=3/2	5-7	3	1	untested	5.0-7.0	-0.0016	0.0001	2	330,388.600
Ne II	ng 2[3] J=5/2	5-7	3	1	untested	5.0-7.0	+0.0010	0.0001	2	330,388.600
Ne II	ng 2[4] J=7/2	5-7	3	1	untested	5.0-7.0	+0.0029	0.0001	2	330,388.600
Ne II	ng 2[5] J=9/2	5-7	3	1	untested	5.0-7.0	+0.0030	0.0001	2	330,388.600
Ne II	ng 2[6] J=11/2	5-7	3	1	untested	5.0-7.0	-0.0002	0.0001	2	330,388.600
Ne II *	nd 2F J=5/2	3-4	2	1	untested	3.0-4.0	+0.0289	0.0055	2	356,229.300
Ne II *	nd 2F J=7/2	3-4	2	1	untested	3.0-4.0	+0.0289	0.0055	2	356,229.300
Ne II *	nd 2G J=7/2	3-4	2	1	untested	2.9-3.9	+0.0674	0.0050	2	356,229.300
Ne II *	nd 2G J=9/2	3-4	2	1	untested	2.9-3.9	+0.0674	0.0050	2	356,229.300
Ne II *	nd 2P J=3/2	3-4	2	1	untested	2.9-3.9	+0.0605	0.0040	2	356,229.300
O III	nd 1D* J=2	3-5	3	1	1/1	2.9-4.9	+0.1196	0.0040	3	443,325.200
O III *	nd 3D* J=1	3-4	2	1	no-triple	2.9-3.9	+0.0791	0.0043	3	443,325.200
O III	nd 3F* J=2	3-5	3	1	1/1	2.9-4.9	+0.1290	0.0090	3	443,325.200
O III *	np 3D J=1	3-4	2	1	no-triple	2.6-3.6	+0.4221	0.0073	3	443,325.200
O III	ns 1P* J=1	3-5	3	1	1/1	2.4-4.4	+0.5852	0.0048	3	443,325.200
O III	ns 3P* J=0	3-5	3	1	1/1	2.4-4.4	+0.6250	0.0048	3	443,325.200
O IV	nd 2D J=3/2	3-6	4	2	2/2	2.9-5.9	+0.0681	0.0025	4	624,202.200
O IV	nf 2F* J=5/2	4-7	4	2	0/2	4.0-6.9	+0.0408	0.0115	4	624,202.200
O IV *	ng 2G J=7/2	5-6	2	1	no-triple	5.0-6.0	-0.0028	0.0018	4	624,202.200
O IV *	nh 2H* J=9/2	6-7	2	1	no-triple	6.0-7.0	-0.0125	0.0030	4	624,202.200
O IV	np 2P* J=1/2	3-5	3	1	1/1	2.7-4.8	+0.2462	0.0106	4	624,202.200
O IV	ns 2S J=1/2	3-5	3	1	1/1	2.6-4.5	+0.4408	0.0072	4	624,202.200
P II *	np 3P J=0	4-5	2	1	no-triple	2.8-3.8	+1.1670	0.0004	2	159,929.200
P II	ns 3P* J=0	4-6	3	1	0/1	2.4-4.4	+1.5468	0.0075	2	159,929.200
P III	nd 2D J=5/2	3-6	4	2	2/2	2.8-5.7	+0.2754	0.0451	3	243,573.700
P III	ng 2G J=7/2	5-8	4	2	1/2	5.0-8.0	+0.0197	0.0011	3	243,573.700
P III	nh 2H* J=9/2	6-8	3	1	1/1	6.0-8.0	+0.0024	0.0012	3	243,573.700
P III	np 2P* J=1/2	4-6	3	1	0/1	3.1-5.1	+0.8886	0.0128	3	243,573.700
P III	ns 2S J=1/2	4-9	6	4	4/4	2.8-7.8	+1.1692	0.0144	3	243,573.700
P IV	nd 1D J=2	4-6	3	1	0/1	3.8-5.8	+0.1991	0.0295	4	415,551.800
P IV	nd 3D J=1	4-6	3	1	0/1	3.8-5.8	+0.2312	0.0239	4	415,551.800
P IV	nf 3F* J=2	4-6	3	1	1/1	4.0-5.9	+0.0597	0.0170	4	415,551.800
P IV	np 1P* J=1	4-6	3	1	0/1	3.3-5.4	+0.6991	0.0588	4	415,551.800
P IV	np 3P* J=0	4-6	3	1	0/1	3.3-5.3	+0.6699	0.0079	4	415,551.800
P IV	ns 1S J=0	4-6	3	1	1/1	3.1-5.1	+0.9018	0.0111	4	415,551.800
P IV	ns 3S J=1	4-7	4	2	0/2	3.1-6.0	+0.9498	0.0073	4	415,551.800
S III *	ns 3P* J=0	4-5	2	1	no-triple	2.7-3.7	+1.2846	0.0049	3	281,130.000
S IV	nd 2D J=3/2	3-5	3	1	1/1	2.8-4.7	+0.2649	0.0197	4	382,135.000
S IV	np 2P* J=1/2	4-7	4	2	1/2	3.2-6.2	+0.7607	0.0167	4	382,135.000
S IV	ns 2S J=1/2	4-7	4	2	1/2	3.0-6.0	+1.0264	0.0100	4	382,135.000
S V	3s.nd 1D J=2	4-6	3	1	0/1	3.8-5.8	+0.1819	0.0275	5	586,610.400
S V	3s.nd 3D J=1	4-7	4	2	1/2	3.8-6.8	+0.2170	0.0211	5	586,610.400
S V	3s.nf 1F* J=3	4-6	3	1	0/1	4.0-6.0	+0.0002	0.0246	5	586,610.400
S V	3s.nf 3F* J=2	4-6	3	1	1/1	4.0-5.9	+0.0637	0.0116	5	586,610.400
S V	3s.np 1P* J=1	4-7	4	2	1/2	3.4-6.4	+0.5900	0.0229	5	586,610.400
S V	3s.np 3P* J=0	4-6	3	1	0/1	3.4-5.3	+0.6221	0.0348	5	586,610.400
S V	3s.ns 1S J=0	4-7	4	2	0/2	3.2-6.2	+0.8043	0.0144	5	586,610.400
S V	3s.ns 3S J=1	4-7	4	2	1/2	3.2-6.1	+0.8428	0.0080	5	586,610.400
S VI	nd 2D J=3/2	3-9†	6	4	2/2	2.9-8.9	+0.0925	0.0069	6	710,194.700
S VI	nf 2F* J=5/2	4-10	7	5	4/5	4.0-10.0	+0.0074	0.0011	6	710,194.700
S VI *	ng 2G J=7/2	5-6	2	1	no-triple	5.0-6.0	+0.0010	0.0001	6	710,194.700
S VI *	nh 2H* J=9/2	6-7	2	1	no-triple	6.0-7.0	+0.0003	0.0001	6	710,194.700
S VI	np 2P* J=1/2	3-7	5	3	3/3	2.6-6.6	+0.4228	0.0110	6	710,194.700
S VI	ns 2S J=1/2	3-8	6	4	4/4	2.4-7.4	+0.6242	0.0083	6	710,194.700
Sc III	3p6.nd 2D J=3/2	3-7	5	3	3/3	2.2-6.4	+0.6538	0.0617	3	199,677.370
Sc III	3p6.nd 2D J=5/2	3-7	5	3	3/3	2.2-6.4	+0.6529	0.0616	3	199,677.370
Sc III	3p6.nf 2F* J=5/2	4-7	4	2	2/2	4.0-6.9	+0.0450	0.0069	3	199,677.370
Sc III	3p6.nf 2F* J=7/2	4-7	4	2	2/2	4.0-6.9	+0.0450	0.0069	3	199,677.370
Sc III	3p6.ng 2G J=7/2	5-8	4	2	2/2	5.0-8.0	+0.0073	0.0007	3	199,677.370
Sc III	3p6.ng 2G J=9/2	5-8	4	2	2/2	5.0-8.0	+0.0073	0.0007	3	199,677.370
Sc III	3p6.np 2P* J=1/2	4-7	4	2	2/2	2.7-5.7	+1.2861	0.0211	3	199,677.370
Sc III	3p6.np 2P* J=3/2	4-7	4	2	2/2	2.7-5.7	+1.2815	0.0211	3	199,677.370
Sc III	3p6.ns 2S J=1/2	4-8	5	3	3/3	2.4-6.4	+1.5848	0.0180	3	199,677.370
Sc III *	3p6.nh 2H* J=11/2	6-7	2	1	no-triple	6.0-7.0	+0.0022	0.0001	3	199,677.370
Sc III *	3p6.nh 2H* J=9/2	6-7	2	1	no-triple	6.0-7.0	+0.0022	0.0001	3	199,677.370
Si I	nd (3/2,3/2)* J=1	20-50	31	29	29/29	20.0-50.0	+0.0126	0.0444	1	66,035.000
Si I	nd (3/2,5/2)* J=3	20-56	37	35	35/35	19.9-56.0	+0.0570	0.0366	1	66,035.000
Si I	ns (3/2,1/2)* J=1	11-24	14	12	untested	9.1-22.1	+1.8546	0.0082	1	66,035.000
Si I	ns (3/2,1/2)* J=2	11-19	9	7	untested	9.1-17.1	+1.8883	0.0005	1	66,035.000
Si I	nf 2[3/2] J=1	4-7	4	2	untested	4.0-7.0	-0.0003	0.0026	1	66,035.000
Si I	nf 2[3/2] J=2	4-7	4	2	untested	4.0-7.0	-0.0008	0.0023	1	66,035.000
Si I	nf 2[5/2] J=2	4-7	4	2	untested	4.0-7.0	+0.0277	0.0056	1	66,035.000
Si I	nf 2[5/2] J=3	4-7	4	2	untested	4.0-7.0	+0.0281	0.0057	1	66,035.000
Si I	nf 2[7/2] J=3	4-7	4	2	untested	4.0-6.9	+0.0471	0.0041	1	66,035.000

The Method 1.6

species	series	n	levels	interior	bracket	n* range	δ	σ(δ)	fits	limit cm ⁻¹
Si I	nf 2[7/2] J=4	4-7	4	2	untested	4.0-6.9	+0.0462	0.0039	1	66,035.000
Si I	nf 2[9/2] J=4	4-7	4	2	untested	4.0-7.0	+0.0165	0.0023	1	66,035.000
Si I	nf 2[9/2] J=5	4-7	4	2	untested	4.0-7.0	+0.0181	0.0027	1	66,035.000
Si I	nd 1D* J=2	3-8	6	4	untested	2.4-7.9	+0.4032	0.1771	1	65,747.760
Si I	nd 1F* J=3	3-8†	5	3	untested	3.0-8.4	-0.1375	0.1513	1	65,747.760
Si I	nd 1P* J=1	3-8	6	4	untested	3.0-8.0	+0.0284	0.0044	1	65,747.760
Si I	nd 3F* J=2	3-8	6	4	untested	2.6-7.6	+0.3730	0.0333	1	65,747.760
Si I	nd 3F* J=3	3-8	6	4	untested	2.6-7.8	+0.3007	0.0615	1	65,747.760
Si I	nd 3F* J=4	3-8	6	4	untested	2.6-8.2	+0.1429	0.2129	1	65,747.760
Si I *	np (1/2,1/2) J=1	6-7	2	1	untested	4.5-5.5	+1.4808	0.0027	1	65,747.760
Si I *	np (1/2,3/2) J=1	6-7	2	1	untested	4.6-5.6	+1.4252	0.0076	1	65,747.760
Si I *	np (1/2,3/2) J=2	6-7	2	1	untested	4.6-5.6	+1.4210	0.0078	1	65,747.760
Si I	ns (1/2,1/2)* J=0	6-10	5	3	3/3	4.1-8.1	+1.8930	0.0038	1	65,747.760
Si I	ns (1/2,1/2)* J=1	6-10	5	3	3/3	4.1-8.1	+1.8758	0.0091	1	65,747.760
Si I	ns (1/2,1/2)* J=0	13-21	9	7	untested	11.1-19.1	+1.8873	0.0031	1	65,747.760
Si I	ns (1/2,1/2)* J=1	13-21	9	7	untested	11.1-19.1	+1.8765	0.0105	1	65,747.760
Si I	nd (1/2,3/2)* J=1	14-44	29	27	27/27	14.0-43.9	+0.0612	0.0310	1	65,747.760
Si I	nd (1/2,3/2)* J=1	9-44†	34	32	untested	9.0-43.9	+0.0584	0.0312	1	65,747.760
Si I	nd (3/2,3/2)* J=0	8-13	6	4	4/4	8.1-13.1	-0.0680	0.0129	1	66,035.000
Si I	nd (3/2,3/2)* J=1	20-50	31	29	untested	20.0-50.0	+0.0129	0.0444	1	66,035.000
Si I	nd (3/2,5/2)* J=1	8-13	6	4	4/4	7.9-12.9	+0.0784	0.0064	1	66,035.000
Si I	nd (3/2,5/2)* J=3	20-56	37	35	untested	19.9-56.0	+0.0573	0.0366	1	66,035.000
Si I	nd (3/2,3/2)* J=0	8-16	9	7	untested	8.1-16.0	-0.0609	0.0146	1	66,035.000
Si I	nd (3/2,5/2)* J=1	8-16	9	7	untested	7.9-15.9	+0.0816	0.0073	1	66,035.000
Si II	3s2.nd 2D J=3/2	3-8	6	4	4/4	2.9-7.7	+0.2290	0.0556	2	131,838.140
Si II	3s2.nd 2D J=5/2	3-8	6	4	4/4	2.9-7.7	+0.2289	0.0558	2	131,838.140
Si II	3s2.nf 2F* J=5/2	4-11	8	6	6/6	3.9-10.9	+0.0846	0.0120	2	131,838.140
Si II	3s2.nf 2F* J=7/2	4-11	8	6	6/6	3.9-10.9	+0.0846	0.0120	2	131,838.140
Si II	3s2.ng 2G J=7/2	5-10	6	4	4/4	5.0-10.0	+0.0174	0.0015	2	131,838.140
Si II	3s2.ng 2G J=9/2	5-10	6	4	4/4	5.0-10.0	+0.0174	0.0015	2	131,838.140
Si II	3s2.np 2P* J=1/2	3-10	8	6	2/6	1.8-9.1	+1.0239	0.0878	2	131,838.140
Si II	3s2.np 2P* J=3/2	3-10	8	6	2/6	1.8-9.1	+1.0215	0.0868	2	131,838.140
Si II	3s2.ns 2S J=1/2	4-9	6	4	4/4	2.6-7.6	+1.3944	0.0161	2	131,838.140
Si III	nd 1D J=2	3-9	7	5	3/5	3.1-9.0	+0.0507	0.1195	3	270,139.300
Si III	nd 3D J=3	3-9	7	5	2/5	2.8-8.8	+0.1977	0.0294	3	270,139.300
Si III	nf 1F* J=3	4-9	6	4	2/4	3.9-9.0	+0.0240	0.1413	3	270,139.300
Si III	nf 3F* J=2	4-9	6	4	4/4	4.0-9.0	+0.0226	0.0272	3	270,139.300
Si III	nf 3F* J=3	4-9	6	4	4/4	4.0-9.0	+0.0224	0.0275	3	270,139.300
Si III	nf 3F* J=4	4-9	6	4	4/4	4.0-9.0	+0.0221	0.0280	3	270,139.300
Si III	ng 1G J=4	5-8	4	2	2/2	5.0-8.0	+0.0266	0.0040	3	270,139.300
Si III	ng 3G J=3	5-9	5	3	3/3	5.0-9.0	+0.0281	0.0047	3	270,139.300
Si III	ng 3G J=4	5-9	5	3	3/3	5.0-9.0	+0.0281	0.0047	3	270,139.300
Si III	ng 3G J=5	5-9	5	3	3/3	5.0-9.0	+0.0280	0.0047	3	270,139.300
Si III	nh 1H* J=5	6-9	4	2	2/2	6.0-9.0	+0.0082	0.0007	3	270,139.300
Si III	nh 3H* J=5	7-9	3	1	1/1	7.0-9.0	+0.0086	0.0004	3	270,139.300
Si III	nh 3H* J=6	6-9	4	2	2/2	6.0-9.0	+0.0082	0.0007	3	270,139.300
Si III	ni 1I J=6	7-9	3	1	1/1	7.0-9.0	+0.0033	0.0002	3	270,139.300
Si III	ni 3I J=7	7-9	3	1	1/1	7.0-9.0	+0.0033	0.0002	3	270,139.300
Si III	np 1P* J=1	4-7	4	2	0/2	3.2-6.3	+0.7576	0.0165	3	270,139.300
Si III	np 3P* J=0	4-7	4	2	2/2	3.2-6.3	+0.7563	0.0149	3	270,139.300
Si III	ns 1S J=0	4-8	5	3	2/3	3.0-7.0	+1.0185	0.0017	3	270,139.300
Si III	ns 3S J=1	4-8	5	3	3/3	2.9-7.0	+1.0646	0.0175	3	270,139.300
Si IV	nd 2D J=5/2	3-8	6	4	4/4	2.9-7.9	+0.0802	0.0079	4	364,093.100
Si IV	nf 2F* J=5/2	4-8	5	3	3/3	4.0-8.0	+0.0054	0.0007	4	364,093.100
Si IV	ng 2G J=7/2	5-8	4	2	2/2	5.0-8.0	+0.0011	0.0001	4	364,093.100
Si IV *	nh 2H* J=9/2	6-7	2	1	no-triple	6.0-7.0	+0.0003	0.0001	4	364,093.100
Si IV	np 2P* J=1/2	3-8	6	4	4/4	2.4-7.5	+0.5262	0.0121	4	364,093.100
Si IV	ns 2S J=1/2	3-8	6	4	4/4	2.2-7.2	+0.7845	0.0094	4	364,093.100
Ti III *	nd 1F J=3	4-5	2	1	no-triple	3.2-4.2	+0.7987	0.0056	3	223,827.100
Ti III *	nd 3D J=1	4-5	2	1	no-triple	3.2-4.2	+0.7899	0.0076	3	223,827.100
Ti III *	np 1D* J=2	4-5	2	1	no-triple	2.6-3.6	+1.4159	0.0063	3	223,827.100
Ti III *	np 3D* J=1	4-5	2	1	no-triple	2.6-3.6	+1.4039	0.0025	3	223,827.100
Ti III *	np 3F* J=2	4-5	2	1	no-triple	2.6-3.6	+1.3956	0.0072	3	223,827.100
Ti III *	ns 1D J=2	4-5	2	1	no-triple	2.3-3.3	+1.6726	0.0013	3	223,827.100
Ti III *	ns 3D J=1	4-5	2	1	no-triple	2.3-3.3	+1.6901	0.0041	3	223,827.100
Ti XI *	nf 1F* J=3	4-5	2	1	no-triple	4.0-5.0	+0.0129	0.0041	11	2,137,900.000
Ti XI	np 1P* J=1	4-7	4	2	1/2	3.6-6.7	+0.3273	0.0161	11	2,137,900.000
Ti XI *	ns 1S J=0	4-5	2	1	no-triple	3.5-4.5	+0.4738	0.0070	11	2,137,900.000
Ti XI *	ns 3S J=1	4-5	2	1	no-triple	3.5-4.5	+0.4983	0.0067	11	2,137,900.000
Zn I	nd 1D J=2	12-20	9	7	untested	10.8-18.8	+1.2232	0.0018	1	75,769.310
Zn I	np 1P* J=1	13-40	28	26	untested	10.9-37.9	+2.0966	0.0033	1	75,769.310
Zn I *	4snd 3D J=1	4-5	2	1	no-triple	2.9-3.9	+1.0940	0.0007	1	75,769.330
Zn I *	4snd 3D J=2	4-5	2	1	no-triple	2.9-3.9	+1.0936	0.0007	1	75,769.330
Zn I *	4snd 3D J=3	4-5	2	1	no-triple	2.9-3.9	+1.0930	0.0007	1	75,769.330
Zn II	ns 2S	4-9	6	4	4/4	1.7-6.8	+2.2084	0.0679	2	144,892.600
Zn II	nd 2D	4-9	6	4	4/4	3.0-8.0	+0.9583	0.0228	2	144,892.600
Zn II	np 2P°	4-8	5	3	3/3	2.1-6.1	+1.8386	0.0616	2	144,892.600
Zn II	nf 2F°	4-8	5	3	3/3	4.0-8.0	+0.0172	0.0044	2	144,892.600
Zn II	ng 2G	5-7	3	1	1/1	5.0-7.0	+0.0038	0.0008	2	144,892.600

596 channel rows across 28 elements · 2,269 interior cells parsed.

element	channels
Al	19
Ar	50
B	32
Ba	27
Be	30
Bi	22
C	36
Ca	18
Cd	12
F	5
Fe	8
Ga	13
Ge	5
He	16
Hg	5
K	8
Li	36
Mg	37
N	18
Na	12
Ne	44
O	12
P	14
S	18
Sc	11
Si	67
Ti	11
Zn	10

III · THE MECHANISMS

How a value is got where no capture reaches. Each is measured on pairs where both cells are known.

mechanism	the move	what it gives
the ground-state derivation	none	the multiplicity, and the bound B
the isoelectronic ladder	$(Z, c) \rightarrow (Z+1, c+1)$	δ along a sequence
the charge ladder	$c \rightarrow c+1$ at fixed Z	δ falls with charge

mechanism	the move	what it gives
ℓ -collapse	$\ell \rightarrow \ell+1$	δ falls with ℓ
the exchange split	singlet $\leftrightarrow$ triplet	the triplet defect exceeds the singlet at ns
the Janet collapse	a lookup in the periodic table	whether a $p = 0$ channel is large
Seaton's formula	none	$\delta_0 = 3\alpha/K(\ell)$ for a non-penetrating series

The isoelectronic ladder

Along a sequence of fixed electron count, a channel's defect follows $\delta(c) = A + B \cdot \ln(c+1)/c$. Fitted per sequence on the 52 with three or more charge states it reaches a median rms of **0.0083** — the tightest relation the compendium holds.

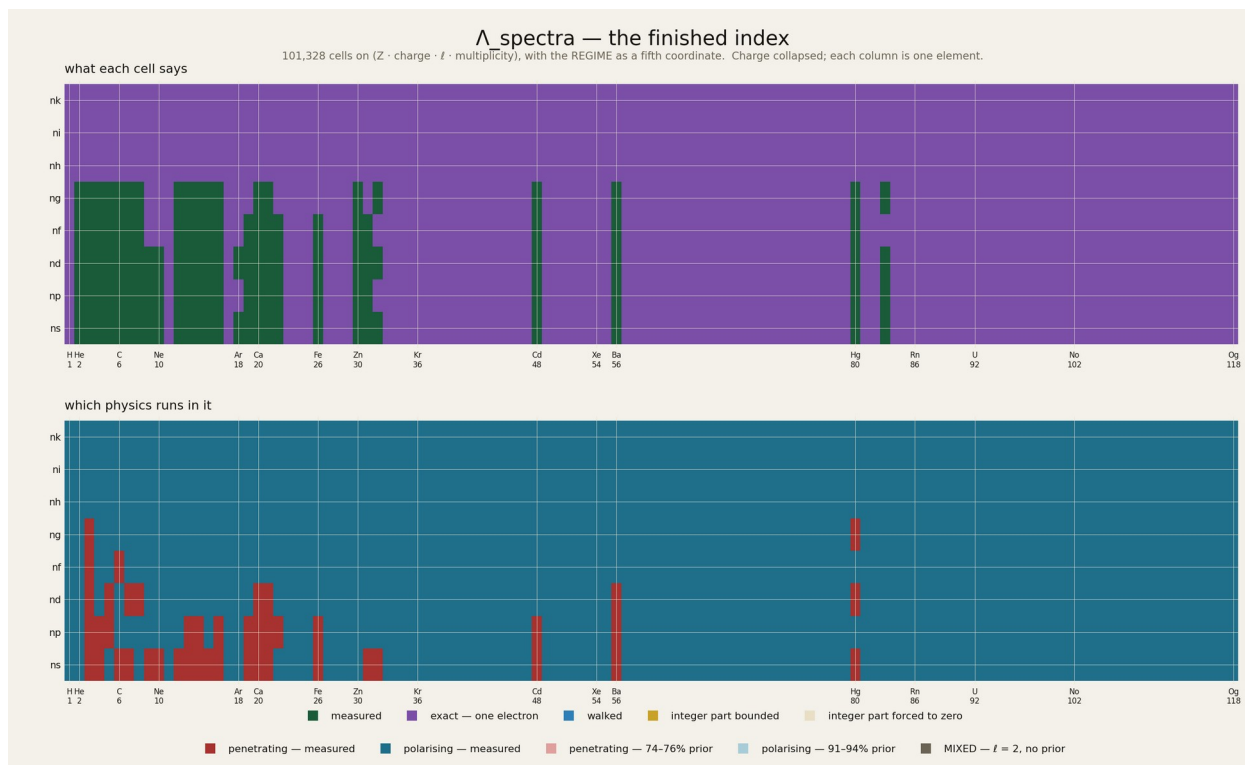
And A is fixed rather than fitted. As charge grows at constant electron count the core collapses to a point and the ion becomes hydrogenic, so $\delta \rightarrow 0$. That is the periodic table's own edge, and setting $A \equiv 0$ leaves one parameter.

A channel is a curve

$$\delta(n) = \delta_0 + \delta_2/(n - \delta_0)^2$$

δ_0 is the limit value the index carries; δ_2 is the curvature. Fitted on 274 channels, the second term removes **48%** of what a constant defect leaves as scatter.

And the sign of δ_2 names the regime: positive for core penetration, negative for core polarisation. By orbital: **s 76% positive, p 74%, d 37%, f 6%, g 7%.** *The sign rule is NIST's own, stated in the reference the levels come from. What the compendium adds is the measurement across 274 channels and the finding that $|\delta_2|$ predicts a channel's scatter at $R^2 = 0.761$.*



The bracket's deficit, on a classical object

The bracket — $T(n)$ between $T(n-1)$ and $T(n+1)$ — is the book's one ternary object (§21.5.1), and Λ_3 is the ternary object of celestial mechanics. The two share the deficit: both need strong 3-consistency and both get level 2 from $\mathcal{R}$. In spectra the deficit is paid per channel by the envelope cells of §12.11.3; in three bodies it is paid once, by the stratum boundaries. No channel entry changes. *The same shortfall, on a classical object, is Λ_3 (Index of Indices V; register 1716, 1724).*

IV · THE SOURCES

B.1 Sources

compilation	spectra drawn
Kaufman & Martin	Al I, Al II
1991, JPCRD 20, 775	
Kramida & Martin	Be I
1997, JPCRD 26, 1185	
NIST ASD	Ar II, Be II, Bi I, C II, Ca II, Cd II, Ga I, He I, He II, Hg II, K II, Li I, Li II, Mg II, N II, Na II, Ne I ($3\frac{1}{2}$), Ne I ($1\frac{1}{2}$), Si I, Si II, Zn II
Sansonetti 2008, JPCRD 37, 1659	Na I

Sansonetti 2008, K I
JPCRD 37, 7

Additional sources used for exotic systems and collective quantities: Hori *et al.*, *Nature* **475**, 484 (2011) and *Phys. Rev. Lett.* **96**, 243401 (2006), both accessed via CODATA 2010 Table XII (arXiv:1203.5425) and arXiv:1304.4330 · Korobov, *Phys. Rev. A* **77**, 042506 (2008) · Singer, Stanojevic, Weidemüller & Côté, *J. Phys. B* **38**, S295 (2005) · Sugar & Corliss 1985, *JPCRD* **14** Suppl. 2 · Sugar & Musgrove 1990, *JPCRD* **19**, 527 and 1995, *JPCRD* **24**, 1803.

B.2 Channels

The channel rows are printed here in full. Each carries its species, channel, n-range, level count, interior cells, bracket, ν range, mean quantum defect, its spread, Z_{eff} and the series limit. *The main volume reproduces one only where it carries an argument; the rest are data, and this is where data belongs.* **THE METHOD 1.6 — SPECTRA COMPENDIUM, counted from the table above rather than from the source that once generated it.** 596 channel rows — 477 series of three or more members and 119 two-member channels (starred) — across 28 elements and 70 species; 3,342 levels, 2,269 interior cells. The bracket stands at **1,577 of 1,738 cells across 392 rows** — the first collection's 844 of 844 on 107 rows; 658 of 813 on 250 rows run under ruling 26 (register 1763); and 75 of 81 on 35 rows run by the same sealed instrument on the six captures the delivering bank's cut carried, each verified against that cut's MANIFEST by hash before a cell was computed (register 1768) — throughout, the sealed test of register 796 at strict interval membership, the only ε the quotation floor — half a unit in the last quoted decimal of the measured level — and admissibility per §22.5, $r = 2Z^2R/(\nu^3\sigma) \geq 5$ with σ the quotation floor. **Cells refused: 0.** The 161 failing cells sit in 98 rows and are results, not defects of the test (register 784: no failures would itself be suspect); five of those rows carry a failing bracket narrower than twice the floor, named in 1763 for the author's eye — none of the six cells failing at this build is among them, each missing a bracket wider than twice its floor. 78 rows read no-triple: their members, matching the row exactly, contain no three consecutive n, so the sealed test — true neighbours only — defines no cell there; the ten added at this build are the ten starred two-member rows of the six delivered species. 126 rows remain untested, and the reason is data, not definition: the parent-term and coupling selection that built them is not reproducible blind (register 1578's wall), and the delivered selection record covers none of their eight species (register 1767). All bracket figures in this paragraph are recomputed from the table above. *The earlier statement of this paragraph — 133 rows, 23 elements, 869 interior cells, bracket 869/869, with a twenty-channel gap disclosed against a total of 153 — was true at registers 630–631 and was overtaken by the parent-term wall (register 1578) and the J-resolved rows of T8-J (registers 1699–1700). It claimed the table was generated and so could not drift; the table did drift, and that generation is retired: the table above is the source, and every count in the paragraph above it is taken from the table — 596, 477, 119, 28, 70, 3,342 and 2,269 agree to the row. The exotic and collective sources named above still carry sources and no rows, and that remains disclosed rather than reconciled.* Registers 630–631; 1578; 1699–1700.

B.3 Flagged channels

Channels whose δ spread exceeds 0.25 or whose V departs from $4\nu/3$ by more than 5% are flagged. Every flag in this collection has an identified cause; none is unexplained.

species	channel	δ spread	cause
Al I	3s ² nd ² D	0.630	compilers relabelled this series; y ² D removed as a perturber
Al II	3snd ¹ D	0.390	local perturber
Al II	3snf ³ F°	0.409	δ reverses sign between n = 6 and 7 — local perturber
Li I	np 2P°	0.678	39 cells to n = 42; quotation coarsens above n = 33
Si II	3s ² np ² P°	0.272	perturber

Published values the coordinates are checked against

source	what it gives
NIST Atomic Spectra Database	the levels every capture is fitted to
Theodosiou, Inokuti & Manson, At. Data Nucl. Data Tables 35, 473 (1986)	asymptotic quantum defects of s, p, d and f orbitals for all ionisation stages of all ions with $Z \leq 50$, Hartree-Slater
Manson, Inokuti & Theodosiou, OSTI 7104964	the isoelectronic, isonuclear and isoionic pictures
Seaton 1958; Drake & Swainson 1991	the polarisation formula
Peper <i>et al.</i> 2019	K I to eight digits
Freeman & Kleppner 1976	Na I ng
arXiv:1706.06237	Cs I np at n = 70-100
arXiv:2508.06733	actinide defects, Z = 89-103
arXiv:2502.20961	Cs ⁺ dipole and quadrupole polarisabilities
ARC Alkali Rydberg Calculator	an independent check on K I

The full bibliography, with what each source was used for, is in the main volume's § References.

V • THE CAPTURE GROUPS

Register 1296 marked these as owed. Counts below are read from the coordinate file set out in § The file, not transcribed.

group	species	cells	measured	exact	computed
Kr-core	Kr	512	0	8	504
Cd/In	Cd/In	1,384	13	16	1,355
La/Ce	La/Ce	1,632	0	16	1,616
Hg-core	Hg	1,168	5	8	1,155

4,696 cells across six species, every one spanning l = 0-7 and the full charge ladder from 1 to Z.

What the counts say, measured on this compendium's own basis

Section 0 states that accuracy is quoted against the WITNESSED-plus-possible set and not against the whole index. The same basis is used here.

The four groups hold 18 reachable cells of the index's 358 - 5.0% of the reachable set on 4.5% of the cells. They are therefore slightly BETTER than representative, not worse.

On the raw grade column they carry 18 measured cells of 4,696 - Cd/In thirteen and Hg five, with **Kr-core and La/Ce carrying none**. *That ratio is not a fact about these groups: the whole index is 98.8% computed, which section 0's table already states. The filler is the index's condition, not these captures' shortcoming.*

Why each was captured

- **Kr-core** - the $n = 4$ closure; its 5s manifold is still one level short
- **Cd/In** - the d10 closure and the first post-d p-electron
- **La/Ce** - THE JANET ANOMALY ITSELF - La opens 5d, Ce is where 4f appears
- **Hg-core** - the 5d closure at high Z, where the relativistic wall stands

La/Ce is the group that earns its place. The other three fill coordinate space; **La/Ce is where the filling order itself breaks**, and holding it in the index lets the anomaly be stated as a coordinate fact rather than a footnote.

VI · THE LÖWDIN SUPPLY

The Löwdin solution is tested against measurement, and this is the measurement it is tested against. Registers 1524–1677 record the capture of these bodies one at a time, each verified before it was written and several corrected after; § Why each was captured does the same office for the coordinate groups. What follows is the empirical supply itself, set out so that a reader can see what was consulted, how much of it there is, and what each body can and cannot carry.

What this supply is, and what it is not

Seven families, 1,816 rows, of which 1,262 were captured from a published compilation and verified before they were held. *Almost none of it is computed by this work. It is the outside world, brought in and checked: X-ray transition energies, ionisation energies along the charge ladder and across the neutral atoms, two level tables, nuclear radii and masses, quantum defects from an independent database, and a staged block of index cells.*

family	rows	span	source compilation
the X-ray lines	474	Ne-Fm, six transitions	NIST SRD 128 v1.2 (Deslattes <i>et al.</i> 2003)
the isoelectronic	546	Z = 1-18 and 19-	NIST ASD ver. 5.12

family	rows	span	source compilation
ladder		36	
the neutral ionisation energies	108	Z = 1-108	NIST ASD ver. 5.12
the level tables	29	Fe I, Kr I	NIST ASD
the nuclear supply	37	Th-Cf, the A = 9 chain	Angeli & Marinova 2013; AME2020
the quantum defects	68	Li, O, Na, Mg, all stages	TOPbase, via de Kertanguy
the staged cells	554	158 index cells	this work's own retrievals
1,816			

The row total and the family total differ, and the difference is the point. *The staged cells are not an outside compilation — they are this work's retrievals staged in the index's own schema, and they are listed here because register 1649 named them as the source of a figure the compendium prints. The outside supply proper is 1,262 rows.*

Two of the eighteen bodies the register names are not held here, and their absence is stated rather than papered over. *The measurement store and the rows recovered at register 652 are stores this work built, not captures it took; register 1674 records that the first has no generator and survives only as data. Neither is an outside compilation, and neither is reproduced in this supply.*

The X-ray transition energies

Six transitions, captured one at a time between registers 1524 and 1555, and the family that cost the most to get right: register 1526 records four fabricated theory values in the first of them, copied from the experimental column where the source stood blank, and the correction is why every later capture was verified by destination address and body header before it was written.

transition	line	rows	first	last
K-L3	K α_1	98	Ne	Fm
L3-M5	L α_1	78	Ca	Fm
L3-M4	L α_2	78	Ca	Fm
L3-M1	L ℓ	88	Na	Fm
L1-N2	L γ_2	67	Ga	Fm
L1-N3	L γ_3	65	As	Fm
474				

Each row carries the element, a mass number where the source gives one, a theoretical energy, a direct experimental energy, and an uncertainty on each. *NIST's compact parenthetical form is preserved as it stands: 849.17(54) is 849.17 $\pm$ 0.54 eV. Two flags travel with the value — an asterisk where the source interpolated from neighbouring elements, and a hash where experiment and theory diverge far enough that the source itself doubts the measurement. A blend field says where K α_1*

and $K\alpha_2$ are unresolved at that element. Blank means blank; nothing is inferred into an empty field.

One row, as it stands:

$K\alpha_1$ U 238 theory 98433.6(36) experiment 98431.58(28) ref 5d

The K-L3 capture carries thirteen actinide isotope rows — uranium at two masses, plutonium at two, americium at two, curium at two, berkelium at two and californium at three — and those are the anchors register 1555 identified. *Californium's third row has no direct experimental value at all; the field is empty in the source and empty here. The mass-number column is blank below bismuth, because the compilation populates it only for radionuclides.*

The physical check the family supports, and it is the reason two of the six were taken. *$L\alpha_2$ must lie below $L\alpha_1$ at every element, since $M4$ is the more tightly bound of the two levels; the pair was captured precisely so that the inequality could be tested rather than assumed. $Ly_3 - Ly_2$ is the $4p$ splitting, and register 1528 took both to see whether the post-krypton offset showed stable structure per region.*

The isoelectronic ladder

Two captures, taken through different endpoints on different days, and their overlap is what makes either trustworthy.

LADDER-K-Kr holds 495 rows: every ionisation stage of every element from potassium to krypton, $Z = 19$ to 36 — the whole 3d block and its shoulders, in electronvolts. *The quality flag is the source's own: 96 measured, 287 estimated, 112 theoretical. That distribution is the ladder's real condition — beyond the first few stages of a mid- Z element, the published value is usually an estimate, and the compendium's high-charge cells inherit that.*

LADDER-H-Ar-I-III holds 51 rows: hydrogen through argon at charges 0, +1 and +2, in wavenumbers, with the ground configuration, the ground level term, the isoelectronic sequence and an uncertainty on each.

One row, as it stands:

H I $Z=1$ charge 0 sequence H ground $1s\ 2S<1/2>$ $109678.77174307\text{ cm}^{-1} \pm 0.00000010$

The two ladders and the neutral table agree where they overlap, and the agreement was tested rather than assumed. *Register 1538 puts eighteen first ionisation energies from a supplied table against the K-Kr capture at a maximum discrepancy of 7.8×10^{-5} eV with nothing beyond 5×10^{-4} ; register 1539 finds the krypton limit agreeing to 4×10^{-9} eV. Three retrievals through three routes, converging — which is corroboration, since the routes are unlike.*

The ionisation energies of the neutral atoms

108 rows, $Z = 1$ to 108 , first ionisation energy of the neutral atom with its uncertainty, and the source's own quality notation carried across intact: 98 measured, 7 estimated, 3 theoretical. *Register 1593 cross-checked these against the K-Kr ladder's own charge-zero rows: eighteen values overlap and all eighteen*

agree exactly, in value and in flag. The square and round brackets the compilation prints are not decoration — they are the quality claim, and dropping them would have discarded the only statement of confidence the source makes.

The two level tables

Fe I — 19 levels across five terms, all five multiplets inverted, as Hund’s third rule requires of a 3d shell more than half filled. Every row passes the wavenumber-to-electronvolt conversion at better than 6×10^{-4} eV. The ionisation potential is excluded, and the a^5P anomaly is flagged in the capture itself rather than in prose alone, so that the flag travels with the data.

Kr I — 10 levels, the $4p^5(^2P_{3/2})5p$ manifold complete at all six. All ten satisfy $J_1\ell$ pair coupling, $K = J_1 \otimes \ell$ and $J = K \pm \frac{1}{2}$. **The table is one row short of the eleven its own header claims, and the missing row is named:** register 1541 settles $85846.709 \text{ cm}^{-1}$ as the $J = 1$ level and records the $J = 0$ metastable as absent from the source, with 86583.760 belonging elsewhere. The capture marks the disputed rows and leaves the choice unmade.

The nuclear supply

23 measured actinide rows — thorium 5, uranium 5, plutonium 6, americium 2, curium 5 — each carrying a change in mean-square charge radius against a reference isotope, an absolute rms charge radius, and an uncertainty on both. From Angeli and Marinova’s 2013 table of experimental ground-state charge radii. Four of the six anchors the argument needs are present and two are absent, which register 1557 predicted from the compilation’s own scope before the table was in hand.

Berkelium and californium are computed, not measured, and the capture says so in its first line. Nine rows, five absolute radii and four differences, from a regression on eighteen pairs across five elements. **They are indicative only and carry their stated error.** The measurement cannot presently be made — register 1559 records that three of the four methods need tens of milligrams of material — so the values are labelled predicted in the file itself, where no later reading can mistake them for a capture.

The $A = 9$ isobar chain — five nuclides, ^9He , ^9Li , ^9Be , ^9B , ^9C — mass excess and uncertainty from AME2020, with a stability note on each. Verified three ways before it was written, including reproduction of the known binding energy per nucleon from the mass excess. **Five nuclides of roughly three and a half thousand:** the capture is a probe of the chain, not a survey of the table, and register 1544 says so in the entry that records it.

The quantum defects of the light ladders

68 cells: lithium 6, oxygen 16, sodium 22, magnesium 24 — each species across every one of its ionisation stages, with the defect given per ℓ . From TOPbase, the Opacity Project’s database, as tabulated by de Kertanguy. The trail was followed there from Theodosiou, Inokuti and Manson’s 1986 compilation, which this work could not obtain directly and which the paper cites as its own reference.

This is the charge-edge extension, and it is the only held source that supplies a complete ladder for a species. Register 1585 identifies it as the most valuable of the

extensions available, for exactly that reason: the compendium's own measured cells cluster at low charge, and a complete ladder is what tests whether the defect's behaviour along the charge axis is what the channel equation says it is.

The staged cells

554 rows over 158 distinct index cells, in the index's own eleven-column schema — the same columns § The file sets out, plus a tier. **Graded A 218, B 157, C 179.** *Many rows to one cell, which is the whole point: register 1662 records that treating repeat measurements of one channel as duplicates inverted the direction of the error, since the second and third measurement of a cell are true measurements with nowhere to sit, not noise.*

Thirty-nine series are staged as skipped, unbounded — no limit stands above them. *A series with no limit above its top member has no defect to compute, and the staging records the fact rather than dropping the row.*

How this supply is cited

A body in this part is cited by its section name, as the coordinate index is cited by COORD(Z, charge, ℓ , $2S+1$). *The register and the volumes both name the bodies. A reader who wants the sodium ladder is sent to § The isoelectronic ladder, not to a filename.*

Nothing in this part is recomputed at press. *The counts are of captures taken and verified on the dates the register records, and they are stated as of capture. § The table and § The file are read from the coordinate companion at build and fail the press if they drift; this part is not, because the bodies it describes are compilations that do not change under this work's hand.*